R_s/R_L

CHAPTER

10 Systems Approach—Effects of R_s and R_L

10.1 INTRODUCTION

In recent years, the introduction of a wide variety of packaged networks and systems has generated an increasing interest in the systems approach to design and analysis. Fundamentally, this approach concentrates on the terminal characteristics of a package and treats each as a building block in the formation of the total package. The content of this chapter is a first step in developing some familiarity with this approach. The techniques introduced will be used in the remaining chapters and broadened as the need arises. The trend to packaged systems is quite understandable when you consider the enormous advances in the design and manufacturing of integrated circuits (ICs). The small IC packages contain stable, reliable, self-testing, sophisticated designs that would be quite bulky if built with discrete (individual) components. The systems approach is not a difficult one to apply once the basic definitions of the various parameters are correctly understood and the manner in which they are utilized is clearly demonstrated. In the next few sections, we develop the systems approach in a slow deliberate manner that will include numerous examples to make each salient point. If the content of this chapter is clearly and correctly understood, a first plateau in the understanding of system analysis will be accomplished.

10.2 TWO-PORT SYSTEMS

The description to follow can be applied to any two-port system—not only those containing BJTs and FETs—although the emphasis in this chapter is on these active devices. The emphasis in previous chapters on determining the two-port parameters for various configurations will be quite helpful in the analysis to follow. In fact, many of the results obtained in the last two chapters are utilized in the analysis to follow.

In Fig. 10.1, the important parameters of a two-port system have been identified. Note in particular the absence of a load and a source resistance. The impact of these important elements is considered in detail in a later section. For the moment recognize that the impedance levels and the gains of Fig. 10.1 are determined for no-load (absence of R_L) and no-source resistance (R_s) conditions.

If we take a "Thévenin look" at the output terminals we find with V_i set to zero that

$$Z_{\text{Th}} = Z_o = R_o \tag{10.1}$$

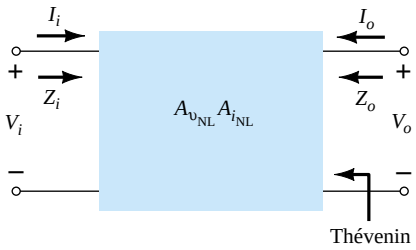


Figure 10.1 Two-port system.

E_{Th} is the open-circuit voltage between the output terminals identified as V_o. However,

$$A_{v_{\text{NL}}} = \frac{V_o}{V_i}$$

and

$$V_o = A_{v_{\text{NL}}} V_i$$

so that

$$\boxed{E_{\text{Th}} = A_{v_{\text{NL}}} V_i} \tag{10.2}$$

Note the use of the additional subscript notation "NL" to identify a no-load voltage gain.

Substituting the Thévenin equivalent circuit between the output terminals will result in the output configuration of Fig. 10.2. For the input circuit the parameters V_i and I_i are related by $Z_i = R_i$, permitting the use of R_i to represent the input circuit. Since our present interest is in BJT and FET amplifiers, both Z_o and Z_i can be represented by resistive elements.

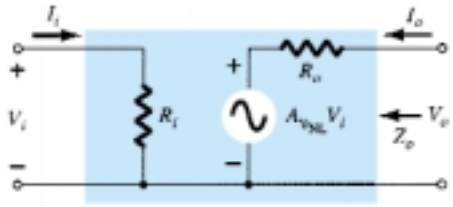


Figure 10.2 Substituting the internal elements for the two-port system of Fig. 10.1.

Before continuing let us check the results of Fig. 10.2 by finding Z_o and $A_{v_{\text{NL}}}$ in the usual manner. To find Z_o, V_i is set to zero, resulting in $A_{v_{\text{NL}}} V_i = 0$, permitting a short-circuit equivalent for the source. The result is an output impedance equal to R_o as originally defined. The absence of a load will result in $I_o = 0$, and the voltage drop across the impedance R_o will be 0 V. The open-circuit output voltage is therefore $A_{v_{\text{NL}}} V_i$, as it should be. Before looking at an example, take note of the fact that A_i does not appear in the two-port model of Fig. 10.2 and in fact is seldom part of the two-port system analysis of active devices. This is not to say that the quantity is seldom calculated, but it is most frequently calculated from the expression $A_i = -A_v(Z_i/R_L)$, where R_L is the defined load for the analysis of interest.

EXAMPLE 10.1

For the fixed-bias transistor network of Fig. 10.3 (Example 8.1), sketch the two-port equivalent of Fig. 10.2.

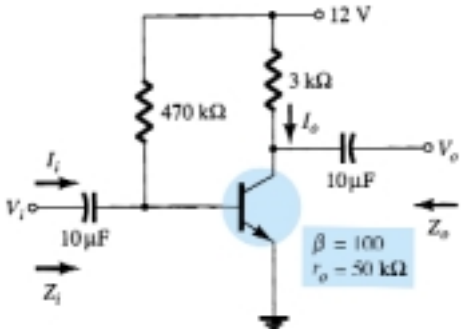


Figure 10.3 Example 10.1.

Solution

From Example 8.1,

$$Z_i = 1.069\ \text{k}\Omega$$
$$Z_o = 3\ \text{k}\Omega$$
$$A_{v_{\text{NL}}} = -280.11$$

Using the information above, the two-port equivalent of Fig. 10.4 can be drawn. Note in particular the negative sign associated with the controlled voltage source, revealing an opposite polarity for the controlled source than that indicated in the figure. It also reveals a 180° phase-shift between the input and output voltages.

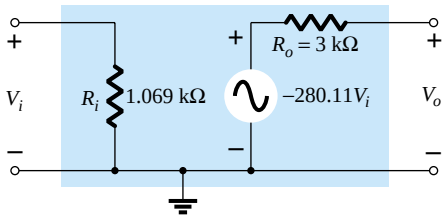


Figure 10.4 Two-port equivalent for the parameters specified in Example 10.1.

In Example 10.1, $R_C = 3\ \text{k}\Omega$ was included in defining the no-load voltage gain. Although this need not be the case (R_C could be defined as the load resistor in Chapter 8), the analysis of this chapter will assume that all biasing resistors are part of the no-load gain and that a loaded system requires an additional load R_L connected to the output terminals.

A second format for Fig. 10.2, particularly popular with op-amps (operational amplifiers), appears in Fig. 10.5. The only change is the general appearance of the model.

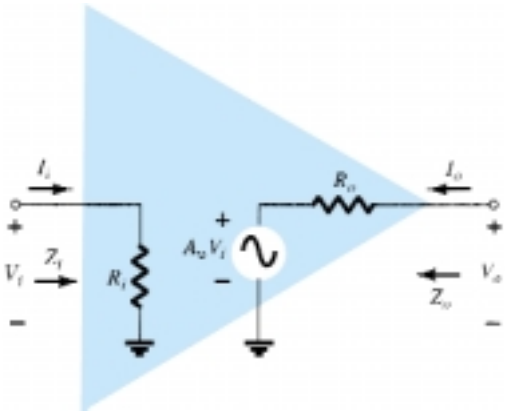


Figure 10.5 Operational amplifier (op-amp) notation.

10.3 EFFECT OF A LOAD IMPEDANCE (R_L)

In this section, the effect of an applied load is investigated using the two-port model of Fig. 10.2. The model can be applied to any current- or voltage-controlled amplifier. $A_{v_{\text{NL}}}$ is, as defined earlier, the gain of the system without an applied load. R_i and R_o are the input and output impedances of the amplifier as defined by the configuration. Ideally, all the parameters of the model are unaffected by changing loads or

source resistances (as normally encountered for op-amps to be described in Chapter 14). However, for some transistor amplifier configurations, R_i can be quite sensitive to the applied load, while for others R_o can be sensitive to the source resistance. In any case, once $A_{v_{NL}}$, R_i, and R_o are defined for a particular configuration, the equations about to be derived can be employed.

Applying a load to the two-port system of Fig. 10.2 will result in the configuration of Fig. 10.6. Applying the voltage-divider rule to the output circuit will result in

$$V_o = \frac{R_L A_{v_{NL}} V_i}{R_L + R_o}$$

and

$$\boxed{A_v = \frac{V_o}{V_i} = \frac{R_L}{R_L + R_o} A_{v_{NL}}} \tag{10.3}$$

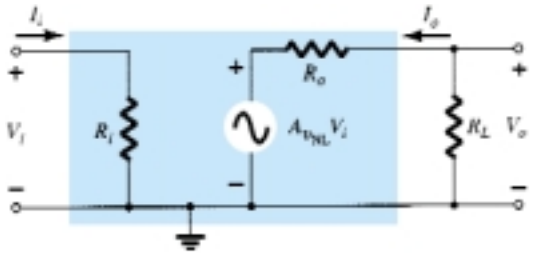


Figure 10.6 Applying a load to the two-port system of Fig. 10.2.

Since the ratio $R_L/(R_L + R_o)$ will always be less than 1:

The loaded voltage gain of an amplifier is always less than the no-load level.

Note also that the formula for the voltage gain does not include the input impedance or current gain.

Although the level of R_i may change with the configuration, the applied voltage and input current will always be related by

$$\boxed{I_i = \frac{V_i}{Z_i} = \frac{V_i}{R_i}} \tag{10.4}$$

Defining the output current as the current through the load will result in

$$\boxed{I_o = -\frac{V_o}{R_L}} \tag{10.5}$$

with the minus sign occurring due to the defined direction for I_o in Fig. 10.6.

The current gain is then determined by

$$A_i = \frac{I_o}{I_i} = \frac{-V_o/R_L}{V_i/Z_i} = -\frac{V_o}{V_i}\frac{Z_i}{R_L}$$

and

$$\boxed{A_i = -A_v\frac{Z_i}{R_L}} \tag{10.6}$$

for the unloaded situation. In general, therefore, the current gain can be obtained from the voltage gain and impedance parameters Z_i and R_L. The next example will demonstrate the usefulness and validity of Eqs. (10.3) through (10.6).

EXAMPLE 10.2

In Fig. 10.7, a load has been applied to the fixed-bias transistor amplifier of Example 10.1 (Fig. 10.3).

(a) Determine the voltage and current gain using the two-port systems approach defined by the model of Fig. 10.4.

(b) Determine the voltage and current gain using the r_e model and compare results.

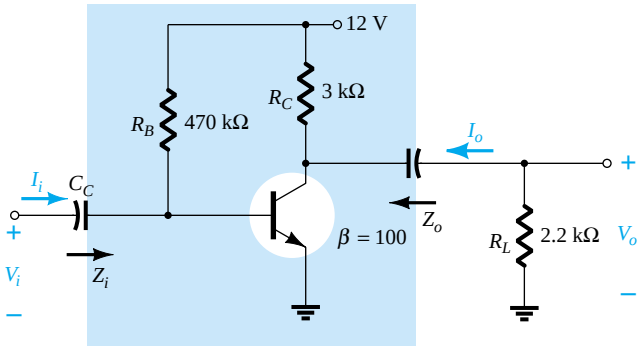


Figure 10.7 Example 10.2.

Solution

(a) Recall from Example 10.1 that

$$Z_i = 1.071\ \text{k}\Omega \qquad \text{(with } r_e = 10.71\ \Omega \text{ and } \beta = 100\text{)}$$

$$Z_o = 3\ \text{k}\Omega$$

$$A_{v_{\text{NL}}} = -280.11$$

Applying Eq. (10.3) yields

$$A_v = \frac{R_L}{R_L + R_o} A_{v_{\text{NL}}}$$

$$= \frac{2.2\ \text{k}\Omega}{2.2\ \text{k}\Omega + 3\ \text{k}\Omega}(-280.11)$$

$$= (0.423)(-280.11)$$

$$= \mathbf{-118.5}$$

For the current gain,

$$A_i = -A_v \frac{Z_i}{R_L}$$

In this case, Z_i is unaffected by the applied load and

$$A_i = -(-118.5)\frac{1.071\ \text{k}\Omega}{2.2\ \text{k}\Omega} = \mathbf{57.69}$$

(b) Substituting the r_e model will result in the network of Fig. 10.8. Note in particular that the applied load is in parallel with the collector resistor R_C defining a net parallel resistance

$$R'_L = R_C \| R_L = 3\ \text{k}\Omega \| 2.2\ \text{k}\Omega = 1.269\ \text{k}\Omega$$

The output voltage

$$V_o = -\beta I_B R'_L$$

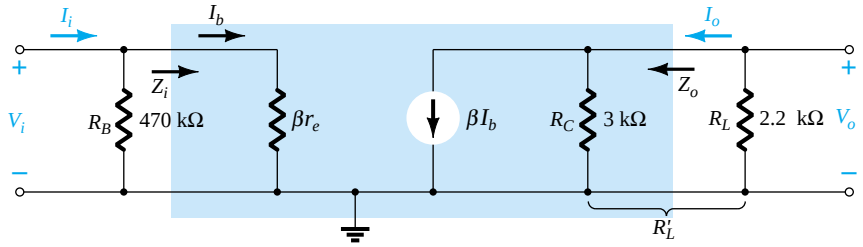


Figure 10.8 Substituting the r_e model in the ac equivalent network of Fig. 10.7.

with
$$I_b = \frac{V_i}{\beta r_e}$$

and
$$V_o = -\beta \frac{V_i}{\beta r_e} R'_L$$

so that
$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{R'_L}{r_e} = -\frac{R_C \| R_L}{r_e}} \qquad (10.7)$$

Substituting values gives

$$A_v = -\frac{1.269\ \text{k}\Omega}{10.71\ \Omega} = \mathbf{-118.5}$$

as obtained above. For the current gain, by the current-divider rule,

$$I_b = \frac{(470\ \text{k}\Omega) I_i}{470\ \text{k}\Omega + 1.071\ \text{k}\Omega} = 0.9977 I_i \cong I_i$$

and
$$I_o = \frac{3\ \text{k}\Omega(\beta I_b)}{3\ \text{k}\Omega + 2.2\ \text{k}\Omega}$$
$$= 0.5769 \beta I_b$$

so that
$$A_i = \frac{I_o}{I_i} = \frac{0.5769 \beta I_b}{I_i} = \frac{0.5769 \beta I_i}{I_i}$$
$$= 0.5769(100) = \mathbf{57.69}$$

as obtained using Eq. (10.6).

Example 10.2 demonstrated two techniques to solve the same problem. Although any network can be solved using the r_e model approach, the advantage of the systems approach is that once the two-port parameters of a system are known, the effect of changing the load can be determined directly from Eq. (10.3). No need to go back to the ac equivalent model and analyze the entire network. The advantages of the systems approach are similar to those associated with applying Thévenin's theorem. They permit concentrating on the effects of the load without having to re-examine the entire network. Of course, if the network of Fig. 10.7 were presented for analysis without the unloaded parameters, it would be a toss-up as to which approach would yield the desired results in the most direct, efficient manner. However, keep in mind that the "package" approach is the developing trend. When you purchase a "system" the two-port parameters are provided, and as with any trend, the user must be aware of how to utilize the given data.

The AC Load Line

For a system such as appearing in Fig. 10.9a, the dc load line was drawn on the output characteristics as shown in Fig. 10.9b. The load resistance did not contribute to the dc load line since it was isolated from the biasing network by the coupling capacitor (C_C). For the ac analysis, the coupling capacitors are replaced by a short-circuit equivalence that will place the load and collector resistors in a parallel arrangement defined by

$$R'_L = R_C \| R_L$$

The effect on the load line is shown in Fig. 10.9b with the levels to determine the new axes intersections. Note of particular importance that the ac and dc load lines pass through the same *Q*-point—a condition that must be satisfied to ensure a common solution for the network under dc and/or ac conditions.

For the unloaded situation, the application of a relatively small sinusoidal signal to the base of the transistor could cause the base current to swing from a level of I_{B_2} to I_{B_4} as shown in Fig. 10.9b. The resulting output voltage v_{ce} would then have the swing appearing in the same figure. The application of the same signal for a loaded situation would result in the same swing in the I_B level, as shown in Fig. 10.9b. The result, however, of the steeper slope of the ac load line is a smaller output voltage

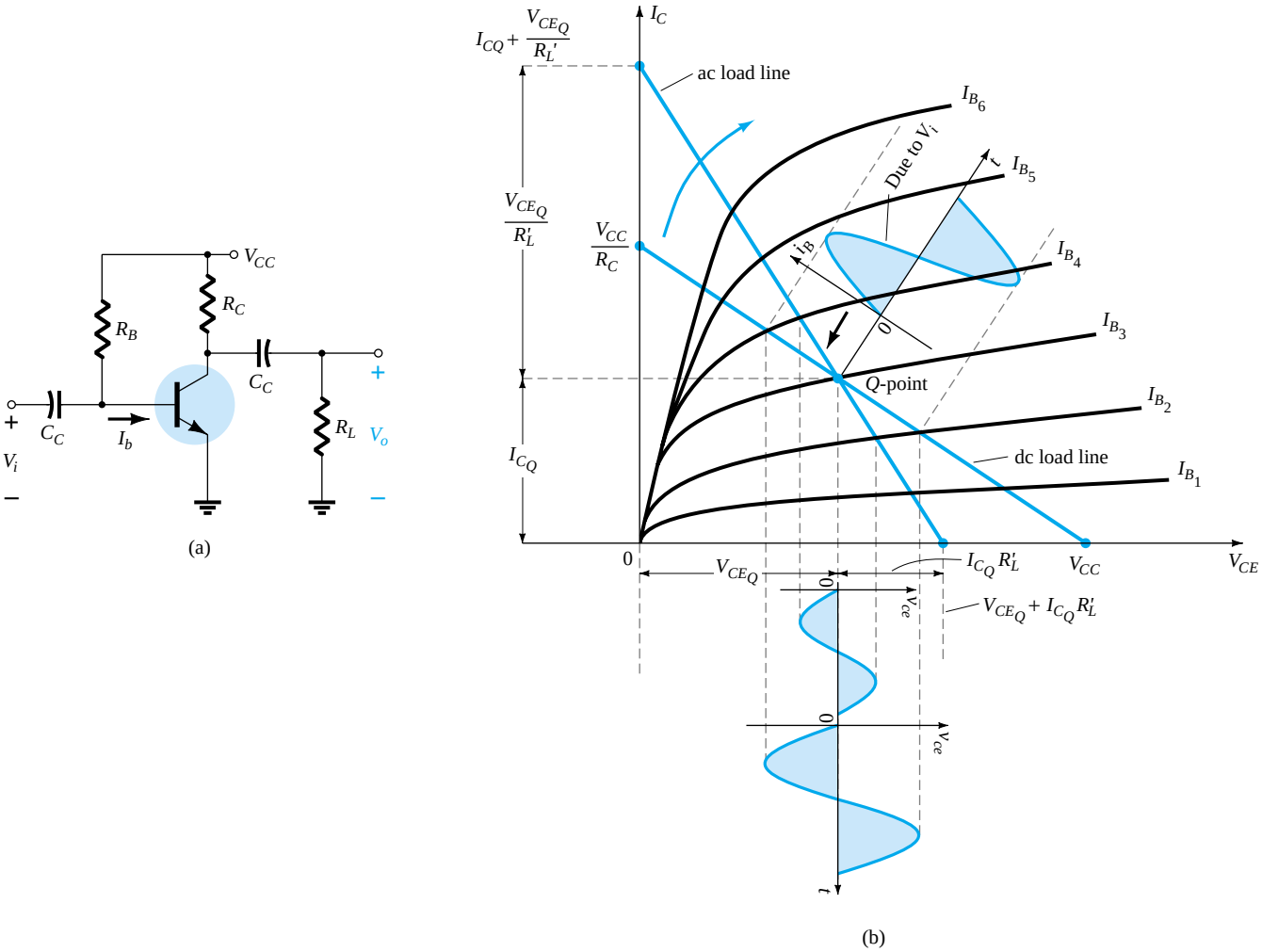


Figure 10.9 Demonstrating the differences between the dc and ac load lines.

swing (v_{ce}) and a drop in the gain of the system as demonstrated in the numerical analysis above. It should be obvious from the intersection of the ac load line on the vertical axis that the smaller the level of R'_L, the steeper the slope and the smaller the ac voltage gain. Since R'_L is smaller for reduced levels of R_L, it should be fairly clear that:

For a particular design, the smaller the level of R_L, the lower the level of ac voltage gain.

10.4 EFFECT OF THE SOURCE IMPEDANCE (R_S)

Our attention will now turn to the input side of the two-port system and the effect of an internal source resistance on the gain of an amplifier. In Fig. 10.10, a source with an internal resistance has been applied to the basic two-port system. The definitions of Z_i and $A_{v_{NL}}$ are such that:

The parameters Z_i and $A_{v_{NL}}$ of a two-port system are unaffected by the internal resistance of the applied source.

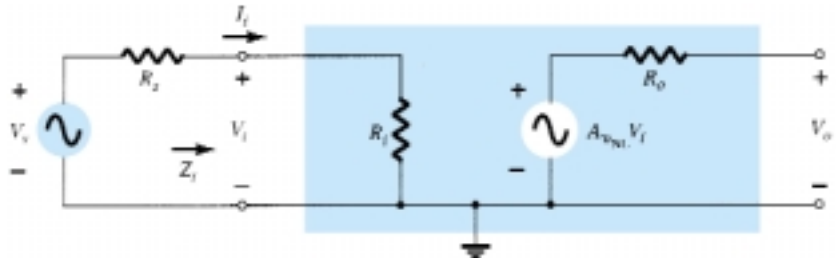


Figure 10.10 Including the effects of the source resistance R_s.

However:

The output impedance may be affected by the magnitude of R_s.

Recall Eq. (8.110) for the complete hybrid equivalent model. The fraction of the applied signal reaching the input terminals of the amplifier of Fig. 10.10 is determined by the voltage-divider rule. That is,

$$V_i = \frac{R_i V_s}{R_i + R_s} \tag{10.8}$$

Equation (10.8) clearly shows that the larger the magnitude of R_s, the less the voltage at the input terminals of the amplifier. In general, therefore:

For a particular amplifier, the larger the internal resistance of a signal source the less the overall gain of the system.

For the two-port system of Fig. 10.10,

$$V_o = A_{v_{NL}} V_i$$

and

$$V_i = \frac{R_i V_s}{R_i + R_s}$$

so that

$$V_o = A_{v_{NL}} \frac{R_i}{R_i + R_s} V_s$$

and

$$A_{v_s} = \frac{V_o}{V_s} = \frac{R_i}{R_i + R_s} A_{v_{NL}} \tag{10.9}$$

The result clearly supports the statement above regarding the reduction in gain with increase in R_s. Using Eq. (10.9), if $R_s = 0\ \Omega$ (ideal voltage source), $A_{v_s} = A_{v_{NL}}$, which is the maximum possible value.

The input current is also altered by the presence of a source resistance as follows:

$$I_i = \frac{V_s}{R_s + R_i} \tag{10.10}$$

EXAMPLE 10.3

In Fig. 10.11, a source with an internal resistance has been applied to the fixed-bias transistor amplifier of Example 10.1 (Fig. 10.3).

(a) Determine the voltage gain $A_{v_s} = V_o/V_s$. What percent of the applied signal appears at the input terminals of the amplifier?

(b) Determine the voltage gain $A_{v_s} = V_o/V_s$ using the r_e model.

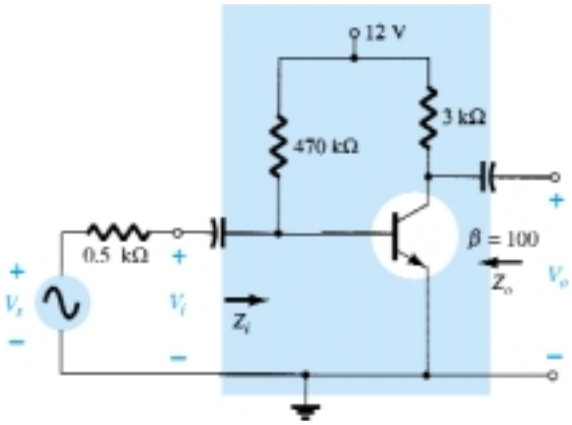


Figure 10.11 Example 10.3.

Solution

(a) The two-port equivalent for the network appears in Fig. 10.12.

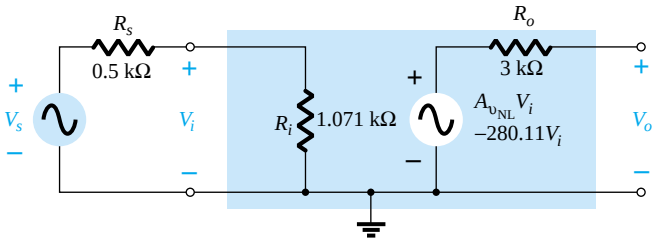


Figure 10.12 Substituting the two-port equivalent network for the fixed-bias transistor amplifier of Fig. 10.11.

$$\text{Eq. (10.9):}\quad A_{v_s} = \frac{V_o}{V_s} = \frac{R_i}{R_i + R_s}A_{v_{NL}} = \frac{1.071\ \text{k}\Omega}{1.071\ \text{k}\Omega + 0.5\ \text{k}\Omega}(-280.11)$$

$$= (0.6817)(-280.11)$$

$$= \mathbf{-190.96}$$

$$\text{Eq. (10.8):}\quad V_i = \frac{R_i V_s}{R_i + R_s} = \frac{(1.071\ \text{k}\Omega)V_s}{1.071\ \text{k}\Omega + 0.5\ \text{k}\Omega} = 0.6817V_s$$

or **68.2%** of the available signal reached the amplifier and 31.8% was lost across the internal resistance of the source.

(b) Substituting the r_e model will result in the equivalent circuit of Fig. 10.13. Solving for V_o gives

$$V_o = -(100I_b)3\ \text{k}\Omega$$

with
$$Z_i \cong \beta r_e \text{ and } I_b \cong I_i = \frac{V_s}{R_s + \beta r_e} = \frac{V_s}{1.571\ \text{k}\Omega}$$

and
$$V_o = -100\left(\frac{V_s}{1.571\ \text{k}\Omega}\right)3\ \text{k}\Omega$$

so that
$$A_{v_s} = \frac{V_o}{V_s} = -\frac{(100)(3\ \text{k}\Omega)}{1.57\ \text{k}\Omega}$$
$$= \mathbf{-190.96}$$

as above.

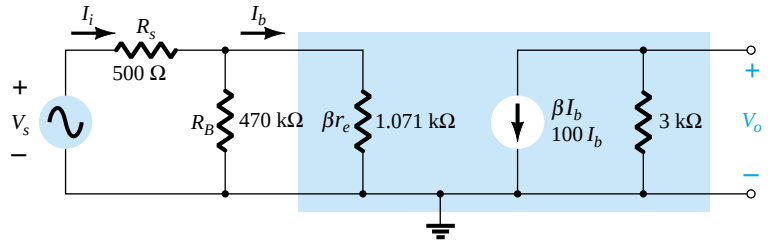


Figure 10.13 Substituting the r_e equivalent circuit for the fixed-bias transistor amplifier of Fig. 10.11.

Throughout the analysis above, note that R_s was not included in the definition of Z_i for the two-port system. Of course, the resistance "seen" by the source is now $R_s + Z_i$, but R_s remains a quantity associated only with the applied source.

Note again in Example 10.3 that the same results were obtained with the systems approach and using the r_e model. Certainly, if the two-port parameters are available, they should be applied. If not, the approach to the solution is simply a matter of preference.

10.5 COMBINED EFFECT OF R_s AND R_L

The effects of R_s and R_L have now been demonstrated on an individual basis. The next natural question is how the presence of both factors in the same network will affect the total gain. In Fig. 10.14, a source with an internal resistance R_s and a load R_L have been applied to a two-port system for which the parameters Z_i, $A_{v_{NL}}$, and Z_o have been specified. For the moment, let us assume that Z_i and Z_o are unaffected by R_L and R_s, respectively.

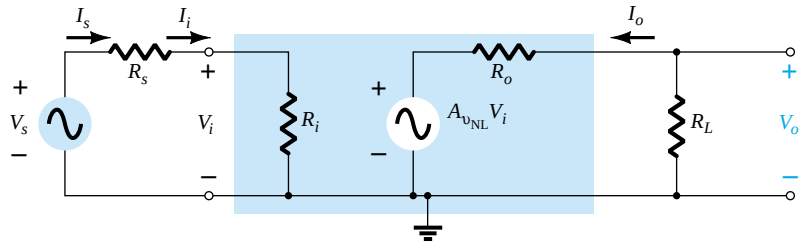


Figure 10.14 Considering the effects of R_s and R_L on the gain of an amplifier.

At the input side we find

$$\text{Eq. (10.8): } V_i = \frac{R_i V_s}{R_i + R_s}$$

or

$$\boxed{\frac{V_i}{V_s} = \frac{R_i}{R_i + R_s}} \tag{10.11}$$

and at the output side,

$$V_o = \frac{R_L A_{v_{NL}} V_i}{R_L + R_o}$$

or

$$\boxed{A_v = \frac{V_o}{V_i} = \frac{R_L A_{v_{NL}}}{R_L + R_o}} \tag{10.12}$$

For the total gain $A_{v_s} = V_o/V_s$, the following mathematical steps can be performed:

$$A_{v_s} = \frac{V_o}{V_s} = \frac{V_o}{V_i}\frac{V_i}{V_s} \tag{10.13}$$

and substituting Eqs. (10.11) and (10.12) will result in

$$A_{v_s} = \frac{R_L A_{v_{NL}}}{R_L + R_o}\frac{R_i}{R_i + R_s}$$

and

$$\boxed{A_{v_s} = \frac{V_o}{V_s} = \frac{R_i}{R_i + R_s}\frac{R_L}{R_L + R_o}A_{v_{NL}}} \tag{10.14}$$

Since $I_i = V_i/R_i$, as before,

$$\boxed{A_i = -A_v\frac{R_i}{R_L}} \tag{10.15}$$

or using $I_s = V_s/(R_s + R_i)$,

$$\boxed{A_{i_s} = -A_{v_s}\frac{R_s + R_i}{R_L}} \tag{10.16}$$

However, $I_i = I_s$, so Eqs. (10.15) and (10.16) will generate the same result. Equation (10.14) clearly reveals that both the source and the load resistance will reduce the overall gain of the system. In fact:

The larger the source resistance and/or smaller the load resistance, the less the overall gain of an amplifier.

The two reduction factors of Eq. (10.14) form a product that has to be carefully considered in any design procedure. It is not sufficient to ensure that R_s is relatively small if the impact of the magnitude of R_L is ignored. For instance, in Eq. (10.14), if the first factor is 0.9 and the second factor is 0.2, the product of the two results in an overall reduction factor equal to (0.9)(0.2) = 0.18, which is close to the lower factor. The effect of the excellent 0.9 level was completely wiped out by the significantly lower second multiplier. If both were 0.9-level factors, the net result would be (0.9)(0.9) = 0.81, which is still quite high. Even if the first were 0.9 and the second 0.7, the net result of 0.63 would still be respectable. In general, therefore, for good overall gain the effect of both R_s and R_L must be evaluated individually and as a product.

EXAMPLE 10.4

For the single-stage amplifier of Fig. 10.15, with $R_L = 4.7\ \text{k}\Omega$ and $R_s = 0.3\ \text{k}\Omega$, determine:

(a) A_{v_s}.

(b) $A_v = V_o/V_i$.

(c) A_i.

The two-port parameters for the fixed-bias configuration are $Z_i = 1.071\ \text{k}\Omega$, $Z_o = 3\ \text{k}\Omega$, and $A_{v_{\text{NL}}} = -280.11$.

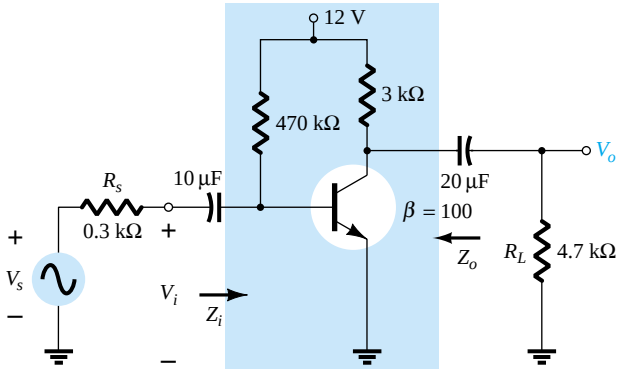


Figure 10.15 Example 10.4

Solution

(a) Eq. (10.14): $A_{v_s} = \dfrac{V_o}{V_s} = \dfrac{R_i}{R_i + R_s}\dfrac{R_L}{R_L + R_o}A_{v_{\text{NL}}}$

$$= \left(\frac{1.071\ \text{k}\Omega}{1.071\ \text{k}\Omega + 0.3\ \text{k}\Omega}\right)\left(\frac{4.7\ \text{k}\Omega}{4.7\ \text{k}\Omega + 3\ \text{k}\Omega}\right)(-280.11)$$

$$= (0.7812)(0.6104)(-280.11)$$

$$= (0.4768)(-280.11)$$

$$= \mathbf{-133.57}$$

(b) $A_v = \dfrac{V_o}{V_i} = \dfrac{R_L A_{v_{\text{NL}}}}{R_L + R_o} = \dfrac{(4.7\ \text{k}\Omega)(-280.11)}{4.7\ \text{k}\Omega + 3\ \text{k}\Omega}$

$$= (0.6104)(-280.11) = \mathbf{-170.98}$$

(c) $A_i = -A_v\dfrac{R_i}{R_L} = -(-170.98)\left(\dfrac{1.071\ \text{k}\Omega}{4.7\ \text{k}\Omega}\right)$

$$= \mathbf{38.96}$$

or $A_{i_s} = -A_{v_s}\dfrac{R_s + R_i}{R_L} = -(-133.57)\left(\dfrac{1.071\ \text{k}\Omega + 0.3\ \text{k}\Omega}{4.7\ \text{k}\Omega}\right)$

$$= \mathbf{38.96}$$

as above.

10.6 BJT CE NETWORKS

The fixed-bias configuration has been employed throughout the analysis of the early sections of this chapter to clearly show the effects of R_s and R_L. In this section, various CE configurations are examined with a load and a source resistance. A detailed analysis will not be performed for each configuration since they follow a very similar path to that demonstrated in the last few sections.

Fixed Bias

For the fixed-bias configuration examined in detail in recent sections, the system model with a load and source resistance will appear as shown in Fig. 10.16. In general,

$$V_o = \frac{R_L}{R_L + R_o} A_{v_{NL}} V_i$$

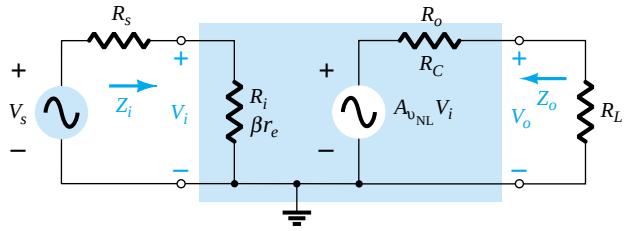


Figure 10.16 Fixed-bias configuration with R_s and R_L

Substituting Eq. (8.6), $A_{v_{NL}} = -R_C/r_e$ and $R_o = R_C$,

$$V_o = -\frac{R_L(-R_C/r_e)V_i}{R_L + R_C}$$

and

$$A_v = \frac{V_o}{V_i} = -\frac{R_L R_C}{R_L + R_C}\frac{1}{r_e}$$

but

$$R_L \| R_C = \frac{R_L R_C}{R_L + R_C}$$

and

$$\boxed{A_v = -\frac{R_L \| R_C}{r_e}} \tag{10.17}$$

If the r_e model were substituted for the transistor in the fixed-bias configuration, the network of Fig. 10.17 would result, clearly revealing that R_C and R_L are in parallel.

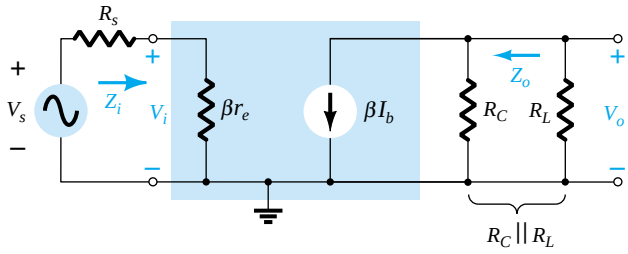


Figure 10.17 Fixed-bias configuration with the substitution of the r_e model.

For the voltage gain A_{v_s} of Fig. 10.16,

$$V_i = \frac{Z_i V_s}{Z_i + R_s}$$

and

$$\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s}$$

with

$$A_{v_s} = \frac{V_o}{V_s} = \frac{V_i}{V_s}\frac{V_o}{V_i}$$

so that
$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_v \tag{10.18}$$

Since the load is connected to the collector terminal of the common-emitter configuration,
$$Z_i = \beta r_e \tag{10.19}$$

and
$$Z_o = R_C \tag{10.20}$$

as obtained earlier.

Voltage-Divider Bias

For the loaded voltage-divider bias configuration of Fig. 10.18, the load is again connected to the collector terminal and Z_i remains
$$Z_i \cong R' \| \beta r_e \qquad (R' = R_1 \| R_2) \tag{10.21}$$

and for the system's output impedance
$$Z_o = R_C \tag{10.22}$$

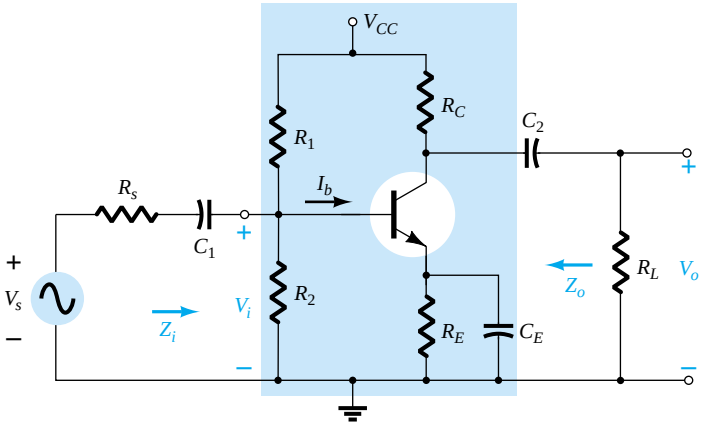


Figure 10.18 Voltage-divider bias configuration with R_s and R_L.

In the small-signal ac model, R_C and R_L will again be in parallel and
$$A_v = -\frac{R_C \| R_L}{r_e} \tag{10.23}$$

with
$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_v \tag{10.24}$$

CE Unbypassed Emitter Bias

For the common-emitter unbypassed emitter-bias configuration of Fig. 10.19, Z_i remains independent of the applied load and
$$Z_i \cong R_B \| \beta R_E \tag{10.25}$$

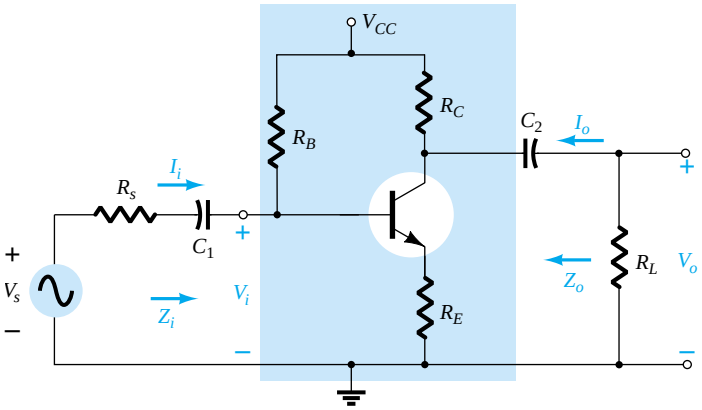


Figure 10.19 CE unbypassed emitter-bias configuration with R_s and R_L.

For the output impedance,

$$Z_o = R_C \tag{10.26}$$

For the voltage gain, the resistance R_C will again drop down in parallel with R_L and

$$A_v = \frac{V_o}{V_i} = -\frac{R_C \| R_L}{R_E} \tag{10.27}$$

with

$$A_{v_s} = \frac{V_o}{V_s} = \frac{Z_i}{Z_i + R_s} A_v \tag{10.28}$$

and

$$A_i = \frac{I_o}{I_i} = -A_v \frac{Z_i}{R_L} \tag{10.29}$$

but keep in mind that $I_i = I_s = V_s/(R_s + Z_i) = V_i/Z_i$.

Collector Feedback

To keep with our connection of the load to the collector terminal the next configuration to be examined is the collector feedback configuration of Fig. 10.20. In the small-signal ac model of the system, R_C and R_L will again drop down in parallel and

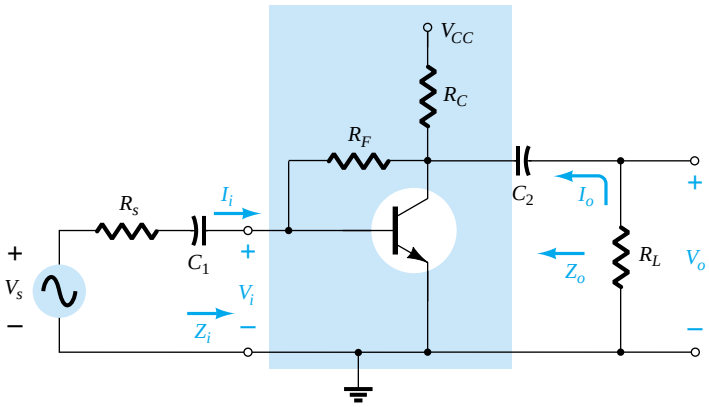


Figure 10.20 Collector feedback configuration with R_s and R_L.

$$A_v = -\frac{R_C \| R_L}{r_e} \tag{10.30}$$

with

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_v \tag{10.31}$$

The output impedance

$$Z_o \cong R_C \| R_F \tag{10.32}$$

and

$$Z_i = \beta r_e \| \frac{R_F}{|A_v|} \tag{10.33}$$

The fact that A_v [Eq. (10.30)] is a function of R_L will alter the level of Z_i from the no-load value. Therefore, if the no-load model is available, the level of Z_i must be modified as demonstrated in the next example.

EXAMPLE 10.5

The collector feedback amplifier of Fig. 10.21 has the following no-load system parameters: $A_{v_{NL}} = -238.94$, $Z_o = R_C \| R_F = 2.66$ kΩ, and $Z_i = 0.553$ kΩ, with $r_e = 11.3$ Ω, and $\beta = 200$. Using the systems approach, determine:

(a) A_v.

(b) A_{v_s}.

(c) A_i.

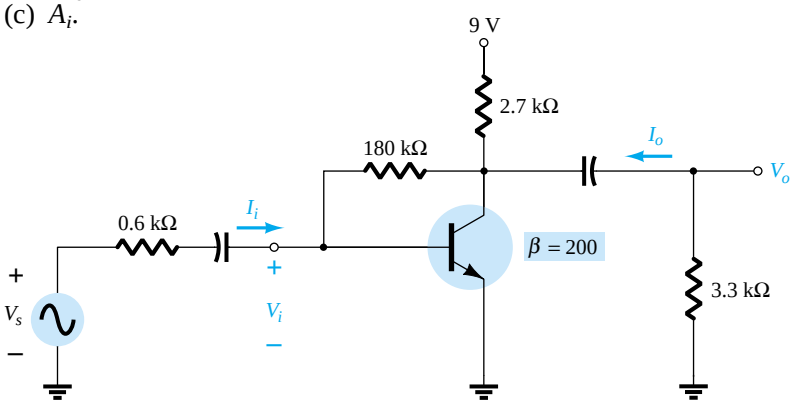


Figure 10.21 Example 10.5.

Solution

(a) For the two-port system:

$$A_v = -\frac{R_C \| R_L}{r_e} = -\frac{2.7\ \text{k}\Omega \| 3.3\ \text{k}\Omega}{11.3\ \Omega}$$

$$= -\frac{1.485\ \text{k}\Omega}{11.3\ \Omega} = \mathbf{-131.42}$$

with

$$Z_i = \beta r_e \| \frac{R_F}{|A_v|} = (200)(11.3\ \Omega) \| \frac{180\ \text{k}\Omega}{131.42}$$

$$= 2.26\ \text{k}\Omega \| 1.37\ \text{k}\Omega$$

$$= 0.853\ \text{k}\Omega$$

The system approach will result in the configuration of Fig. 10.22 with the value of Z_i as controlled by R_L and the voltage gain. Now the two-port gain equation can be applied (slight difference in A_v due to approximation $\beta I_b \gg I_{R_F}$ in Section 8.7):

$$A_v = \frac{R_L A_{v_{NL}}}{R_L + R_o} = \frac{(3.3\ \text{k}\Omega)(-238.94)}{3.3\ \text{k}\Omega + 2.66\ \text{k}\Omega} = \mathbf{-132.3}$$

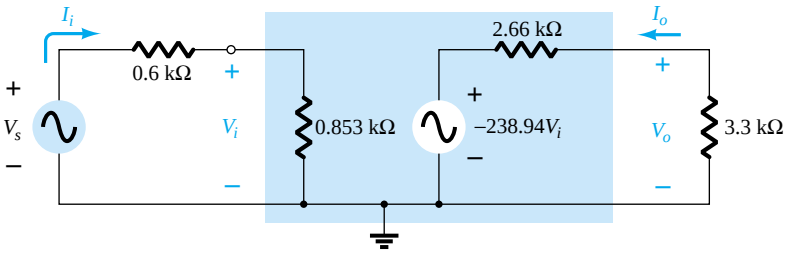


Figure 10.22 The ac equivalent circuit for the network of Fig. 10.21.

(b) $$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_v = \frac{0.853\ \text{k}\Omega}{0.853\ \text{k}\Omega + 0.6\ \text{k}\Omega}(-132.3)$$
$$= \mathbf{-77.67}$$

(c) $$A_i = -A_v \frac{Z_i}{R_L} = -(-132.3)\left(\frac{0.853\ \text{k}\Omega}{3.3\ \text{k}\Omega}\right) = \frac{(132.3)(0.853\ \text{k}\Omega)}{3.3\ \text{k}\Omega}$$
$$= \mathbf{34.2}$$

or $$A_i = -A_{v_s} \frac{Z_i + R_s}{R_L} = -(-77.67)\left(\frac{0.853\ \text{k}\Omega + 0.6\ \text{k}\Omega}{3.3\ \text{k}\Omega}\right)$$
$$= \mathbf{34.2}$$

10.7 BJT EMITTER-FOLLOWER NETWORKS

The input and output impedance parameters of the two-port model for the emitter-follower network are sensitive to the applied load and source resistance. For the emitter-follower configuration of Fig. 10.23, the small-signal ac model would appear as shown in Fig. 10.24. For the input section of Fig. 10.24, the resistance R_B is neglected because it is usually so much larger than the source resistance that a Thévenin equiv-

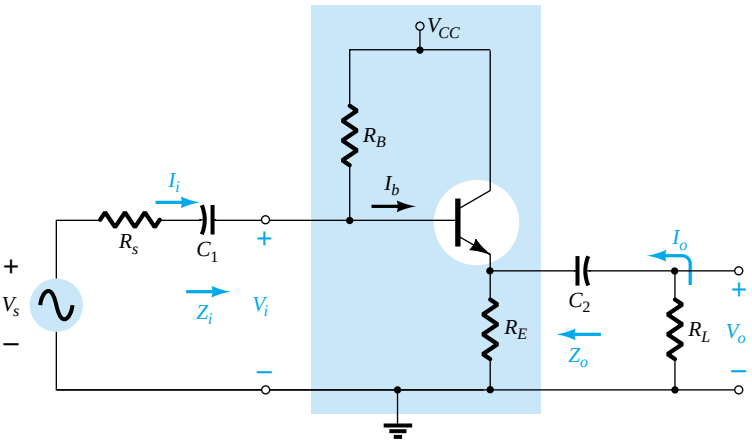


Figure 10.23 Emitter-follower configuraton with R_s and R_L.

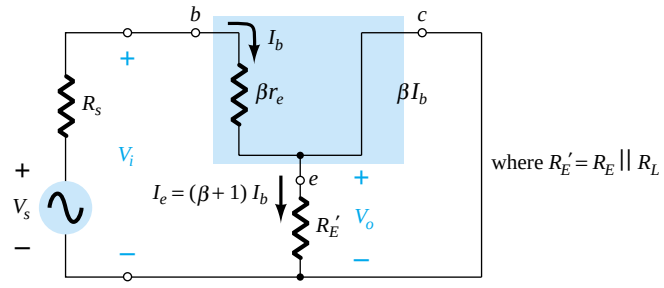


Figure 10.24 Emitter-follower configuration of Fig. 10.23 following the substitution of the r_e equivalent circuit.

alent circuit for the configuration of Fig. 10.25 would result in simply R_s and V_s as shown in Fig. 10.24. Of course, if current levels are to be determined such as I_i in the original diagram, the effect of R_B must be included.

Applying Kirchhoff's voltage law to the input circuit of Fig. 10.24 will result in

$$V_s - I_b R_s - I_b \beta r_e - (\beta + 1) I_b R_E' = 0$$

and

$$V_s - I_b(R_s + \beta r_e + (\beta + 1) R_E') = 0$$

so that

$$I_b = \frac{V_s}{R_s + \beta r_e + (\beta + 1) R_E'}$$

Establishing I_e, we have

$$I_e = (\beta + 1) I_b = \frac{(\beta + 1) V_s}{R_s + \beta r_e + (\beta + 1) R_E'}$$

and

$$I_e = \frac{V_s}{[(R_s + \beta r_e)/(\beta + 1)] + R_E'}$$

Using $\beta + 1 \cong \beta$ yields

$$\boxed{I_e = \frac{V_s}{(R_s/\beta + r_e) + R_E'}} \tag{10.34}$$

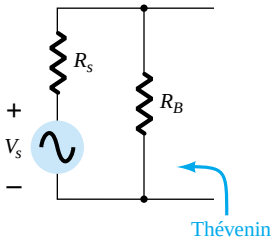


Figure 10.25 Determining the Thévenin equivalent circuit for the input circuit of Fig. 10.23.

Drawing the network to "fit" Eq. (10.34) will result in the configuration of Fig. 10.26a. In Fig. 10.26b, R_E and the load resistance R_L have been separated to permit a definition of Z_o and I_o.

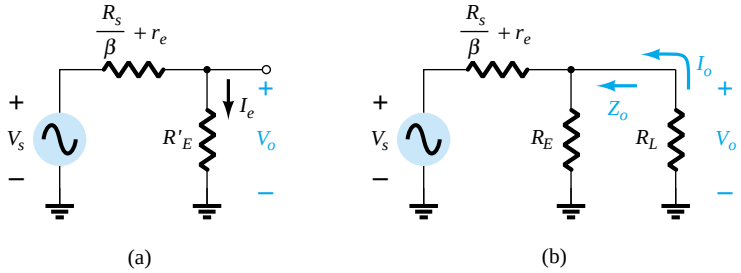


Figure 10.26 Networks resulting from the application of Kirchhoff's voltage law to the input circuit of Fig. 10.24.

The voltage gain can then be obtained directly from Fig. 10.26a using the voltage divider rule:

$$V_o = \frac{R_E' V_s}{R_E' + (R_s/\beta + r_e)}$$

or

$$A_{v_s} = \frac{V_o}{V_s} = \frac{R_E'}{R_E' + (R_s/\beta + r_e)}$$

and
$$A_{v_s} = \frac{V_o}{V_s} = \frac{R_E \| R_L}{R_E \| R_L + R_s/\beta + r_e} \tag{10.35}$$

Setting $V_s = 0$ and solving for Z_o will result in

$$Z_o = R_E \| \left(\frac{R_s}{\beta} + r_e \right) \tag{10.36}$$

For the input impedance,

$$Z_b = \beta(r_e + R'_E)$$

and
$$Z_i = R_B \| Z_b$$

or
$$Z_i = R_B \| \beta(r_e + R_E \| R_L) \tag{10.37}$$

For no-load conditions, the gain equation is

$$A_{v_{NL}} \cong \frac{R_E}{R_E + r_e}$$

while for loaded conditions,

$$A_v \cong \frac{V_o}{V_i} = \frac{R_E \| R_L}{R_E \| R_L + r_e} \tag{10.38}$$

EXAMPLE 10.6

For the loaded emitter-follower configuration of Fig. 10.27 with a source resistance and the no-load two-port parameters of $Z_i = 157.54\ \text{k}\Omega$, $Z_o = 21.6\ \Omega$, and $A_{v_{NL}} = 0.993$ with $r_e = 21.74\ \Omega$ and $\beta = 65$, determine:

(a) The new values of Z_i and Z_o as determined by the load and R_s, respectively.
(b) A_v using the systems approach.
(c) A_{v_s} using the systems approach.
(d) $A_i = I_o/I_i$.

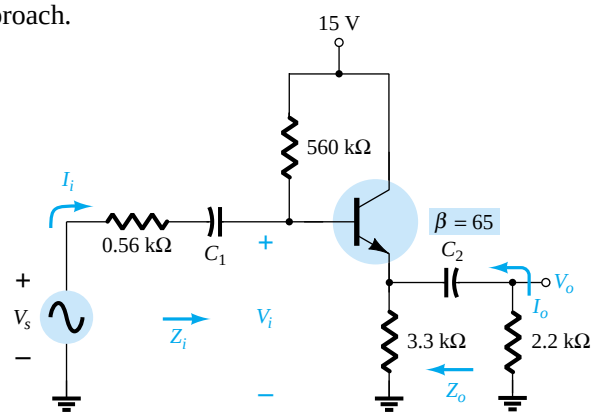


Figure 10.27 Example 10.6.

Solution

Eq. (10.37): $Z_i = R_B \| \beta(r_e + R_E \| R_L)$

$= 560\ \text{k}\Omega \| 65(21.74\ \Omega + \underbrace{3.3\ \text{k}\Omega \| 2.2\ \text{k}\Omega}_{1.32\ \text{k}\Omega})$

$= 560\ \text{k}\Omega \| 87.21\ \text{k}\Omega$

$= \mathbf{75.46\ k\Omega}$

versus 157.54 kΩ (no-load).

$$Z_o = R_E \| \left(\frac{R_s}{\beta} + r_e \right)$$

$$= 3.3\ \text{k}\Omega \| \left(\frac{0.56\ \text{k}\Omega}{65} + 21.74\ \Omega \right)$$

$$= 3.3\ \text{k}\Omega \| 30.36\ \Omega$$

$$= \mathbf{30.08\ \Omega}$$

versus 21.6 Ω (no R_s).

(b) Substituting the two-port equivalent network will result in the small-signal ac equivalent network of Fig. 10.28.

$$V_o = \frac{R_L A_{v_{\text{NL}}} V_i}{R_L + R_o} = \frac{(2.2\ \text{k}\Omega)(0.993)V_i}{2.2\ \text{k}\Omega + 30.08\ \Omega}$$

$$\cong 0.98 V_i$$

$$\text{with } A_v = \frac{V_o}{V_i} \cong \mathbf{0.98}$$

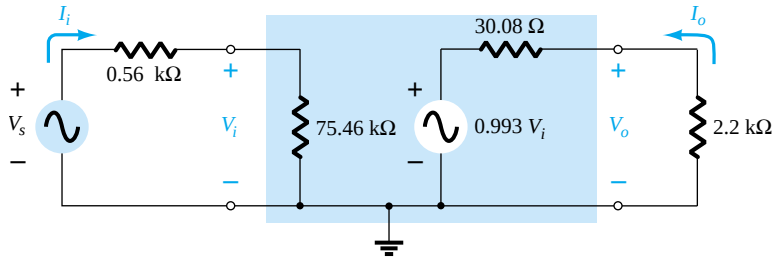


Figure 10.28 Small-signal ac equivalent circuit for the network of Fig. 10.27.

(c) $V_i = \dfrac{Z_i V_s}{Z_i + R_s} = \dfrac{(75.46\ \text{k}\Omega)V_s}{75.46\ \text{k}\Omega + 0.56\ \text{k}\Omega} = 0.993 V_s$

so that $$A_{v_s} = \frac{V_o}{V_s} = \frac{V_o}{V_i}\frac{V_i}{V_s} = (0.98)(0.993) = \mathbf{0.973}$$

(d) $A_i = \dfrac{I_o}{I_i} = -A_v \dfrac{Z_i}{R_L}$

$$= -(0.98)\left(\frac{75.46\ \text{k}\Omega}{2.2\ \text{k}\Omega} \right)$$

$$= \mathbf{-33.61}$$

10.8 BJT CB NETWORKS

A common-base amplifier with an applied load and source resistance appear in Fig. 10.29. The fact that the load is connected between the collector and base terminals isolates it from the input circuit, and Z_i remains essentially the same for no-load or loaded conditions. The isolation that exists between input and output circuits also maintains Z_o at a fixed level even though the level of R_s may change. The voltage gain is now determined by

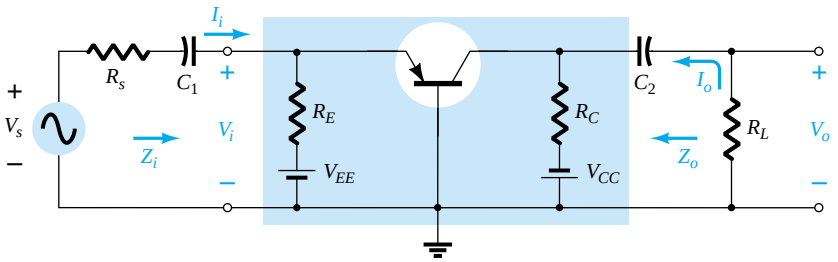


Figure 10.29 Common-base configuration with R_s and R_L.

$$A_v \cong \frac{R_C \| R_L}{r_e} \tag{10.39}$$

and the current gain:

$$A_i \cong -1 \tag{10.40}$$

EXAMPLE 10.7

For the common-base amplifier of Fig. 10.30, the no-load two-port parameters are (using $\alpha \cong 1$) $Z_i \cong r_e = 20\ \Omega$, $A_{v_{NL}} = 250$, and $Z_o = 5\ \text{k}\Omega$. Using the two-port equivalent model, determine:

(a) A_v.
(b) A_{v_s}.
(c) A_i.

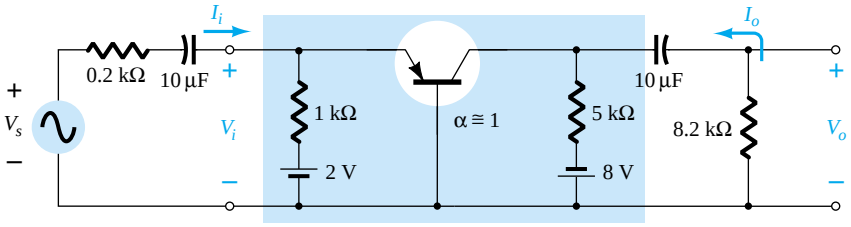


Figure 10.30 Example 10.7.

Solution

(a) The small-signal ac equivalent network appears in Fig. 10.31.

$$V_o = \frac{R_L A_{v_{NL}} V_i}{R_L + R_o} = \frac{(8.2\ \text{k}\Omega)(250)V_i}{8.2\ \text{k}\Omega + 5\ \text{k}\Omega} = 155.3V_i$$

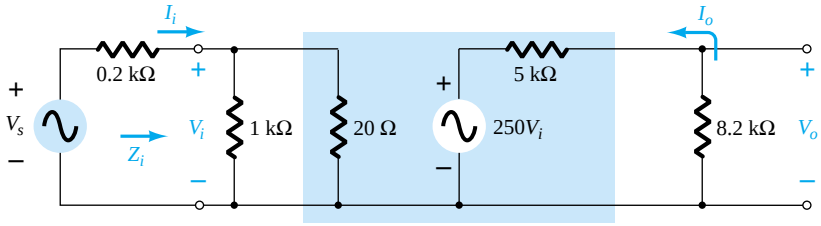


Figure 10.31 Small-signal ac equivalent circuit for the network of Fig. 10.30.

and $$A_v = \frac{V_o}{V_i} = \mathbf{155.3}$$

or $$A_v \cong \frac{R_C \| R_L}{r_e} = \frac{5\ \text{k}\Omega \| 8.2\ \text{k}\Omega}{20\ \Omega} = \frac{3.106\ \text{k}\Omega}{20\ \Omega}$$
$$= \mathbf{155.3}$$

(b) $$A_{v_s} = \frac{V_o}{V_s} = \frac{V_i}{V_s}\frac{V_o}{V_i}$$
$$= \frac{R_i}{R_i + R_s} A_v = \left(\frac{20\ \Omega}{20\ \Omega + 200\ \Omega}\right)(155.3)$$
$$= \mathbf{14.12}$$

Note the relatively low gain due to a source impedance much larger than the input impedance of the amplifier.

(c) $$A_i = -A_v \frac{Z_i}{R_L} = -(155.3)\left(\frac{20\ \Omega}{8.2\ \text{k}\Omega}\right)$$
$$= \mathbf{-0.379}$$

which is significantly less than 1 due to the division of output current between R_C and R_L.

10.9 FET NETWORKS

As noted in Chapter 9, the isolation that exists between gate and drain or source of an FET amplifier ensures that changes in R_L do not affect the level of Z_i and changes in R_{sig} do not affect R_o. In essence, therefore:

The no-load two-port model of Fig. 10.2 for an FET amplifier is unaffected by an applied load or source resistance.

Bypassed Source Resistance

For the FET amplifier of Fig. 10.32, the applied load will appear in parallel with R_D in the small-signal model, resulting in the following equation for the loaded gain:

$$A_v = -g_m(R_D \| R_L) \tag{10.41}$$

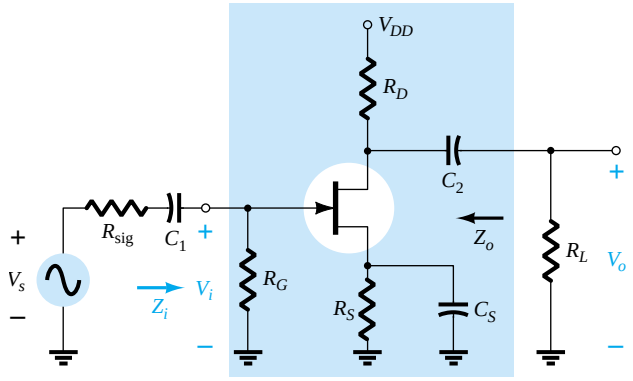


Figure 10.32 JFET amplifier with R_{sig} and R_L.

The impedance levels remain at

$$Z_i = R_G \tag{10.42}$$

$$Z_o = R_D \tag{10.43}$$

Unbypassed Source Resistance

For the FET amplifier of Fig. 10.33, the load will again appear in parallel with R_D and the loaded gain becomes

$$A_v = \frac{V_o}{V_i} = -\frac{g_m(R_D \| R_L)}{1 + g_m R_S} \tag{10.44}$$

with

$$Z_i = R_G \tag{10.45}$$

and

$$Z_o = R_D \tag{10.46}$$

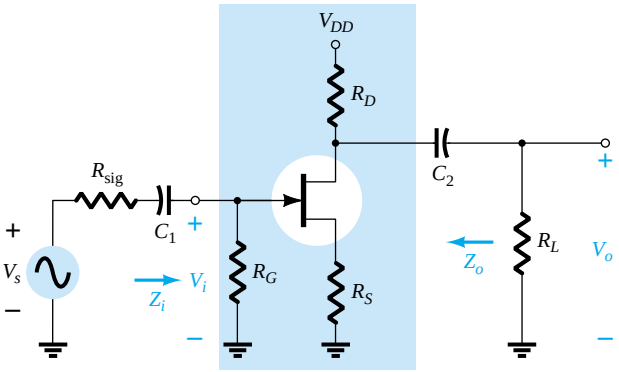


Figure 10.33 JFET amplifier with unbypassed R_S.

EXAMPLE 10.8

For the FET amplifier of Fig. 10.34, the no-load two-port parameters are $A_{v_{NL}} = -3.18$, $Z_i = R_1 \| R_2 = 239$ kΩ, and $Z_o = 2.4$ kΩ, with $g_m = 2.2$ mS.

(a) Using the two-port parameters above, determine A_v and A_{v_s}.

(b) Using Eq. (10.44), calculate the loaded gain and compare to the result of part (a).

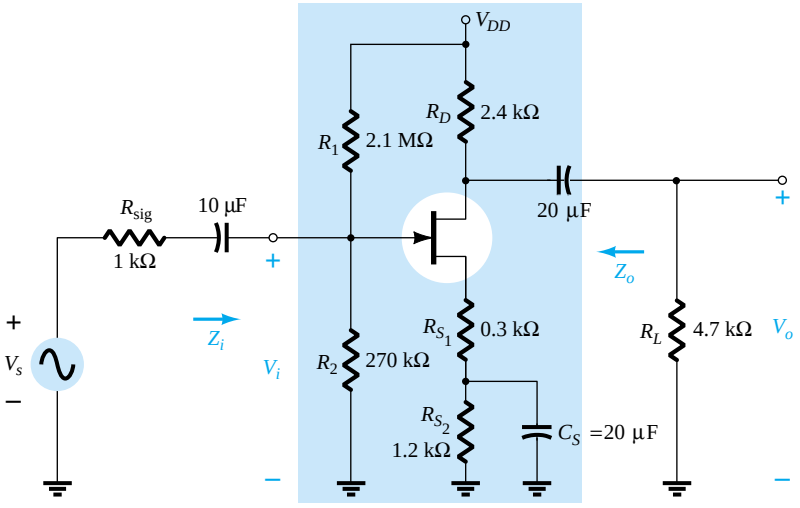


Figure 10.34 Example 10.8.

Solution

(a) The small-signal ac equivalent network appears in Fig. 10.35, and

$$A_v = \frac{V_o}{V_i} = \frac{R_L A_{v_{NL}}}{R_L + R_o} = \frac{(4.7\ \text{k}\Omega)(-3.18)}{4.7\ \text{k}\Omega + 2.4\ \text{k}\Omega}$$

$$= \mathbf{-2.105}$$

$$A_{v_s} = \frac{V_o}{V_s} = \frac{V_i}{V_s}\frac{V_o}{V_i} = \frac{R_i}{R_i + R_{\text{sig}}} A_v$$

$$= \frac{(239\ \text{k}\Omega)(-2.105)}{239\ \text{k}\Omega + 1\ \text{k}\Omega}$$

$$= \mathbf{-2.096} \cong A_v$$

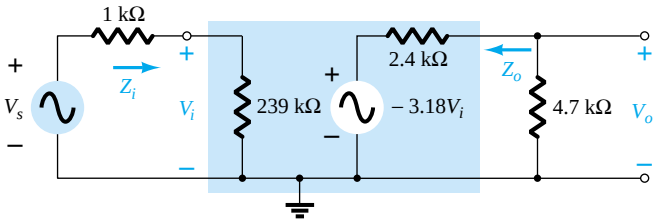


Figure 10.35 Small-signal ac equivalent circuit for the network of Fig. 10.34.

(b) Eq. (10.44): $A_v = \dfrac{-g_m(R_D \| R_L)}{1 + g_m R_{S_1}}$

$$= \frac{-(2.2\ \text{mS})(2.4\ \text{k}\Omega \| 4.7\ \text{k}\Omega)}{1 + (2.2\ \text{mS})(0.3\ \text{k}\Omega)} = \frac{-3.498}{1.66}$$

$$= \mathbf{-2.105}\ \text{as above}$$

Source Follower

For the source-follower configuration of Fig. 10.36, the level of Z_i is independent of the magnitude of R_L and determined by

$$\boxed{Z_i = R_G} \qquad (10.47)$$

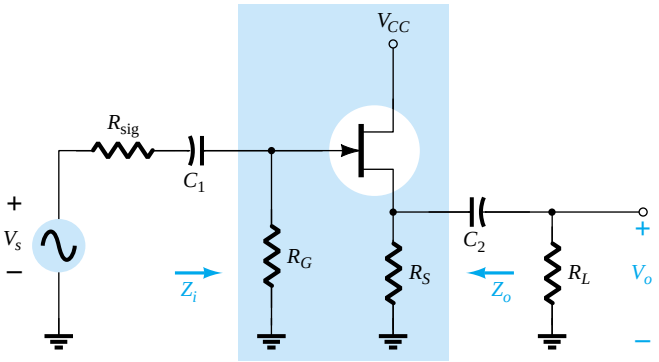


Figure 10.36 Source-follower configuration with R_{sig} and R_L.

The loaded voltage gain has the same format as the unloaded gain with R_S replaced by the parallel combination of R_S and R_L.

$$A_v = \frac{V_o}{V_i} = \frac{g_m(R_S \| R_L)}{1 + g_m(R_S \| R_L)} \tag{10.48}$$

The level of output impedance is as determined in Chapter 9:

$$Z_o = R_S \| \frac{1}{g_m} \tag{10.49}$$

revealing an insensitivity to the magnitude of the source resistance R_{sig}.

Common Gate

Even though the common-gate configuration of Fig. 10.37 is somewhat different from those described above with regard to the placement of R_L and R_{sig}, the input and output circuits remain isolated and

$$Z_i = \frac{R_S}{1 + g_m R_S} \tag{10.50}$$

$$Z_o = R_D \tag{10.51}$$

The loaded voltage gain is given by

$$A_v = g_m(R_D \| R_L) \tag{10.52}$$

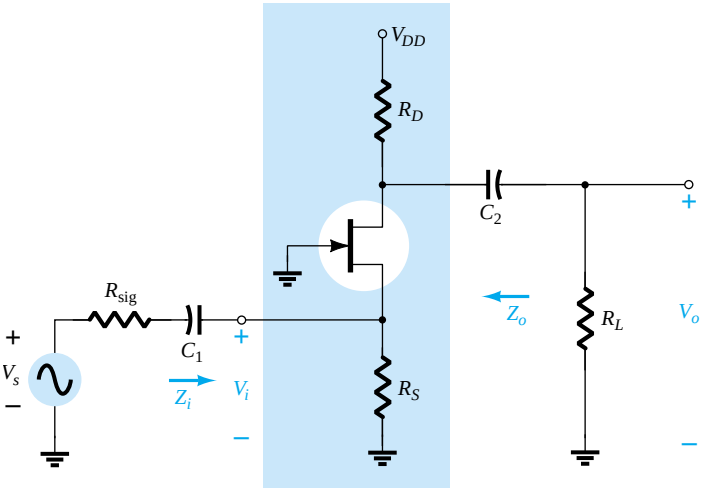


Figure 10.37 Common-gate configuration with R_{sig} and R_L.

10.10 SUMMARY TABLE

Now that the loaded and unloaded (Chapters 8 and 9) BJT and JFET amplifiers have been examined in some detail, a review of the equations developed is provided by Table 10.1. Although all the equations are for the loaded situation, the removal of R_L will result in the equations for the unloaded amplifier. The same is true for the effect of R_s (for BJTs) and R_{sig} (for JFETs) on Z_o. In each case, the phase relationship between the input and output voltages is also provided for quick reference. A review of

the equations will reveal that the isolation provided by the JFET between the gate and channel by the SiO_2 layer results in a series of less complex equations than those encountered for the BJT configurations. The linkage provided by I_b between input and output circuits of the BJT transistor amplifier adds a touch of complexity to some of the equations.

TABLE 10.1 Summary of Transistor Configurations (A_v, Z_i, Z_o)

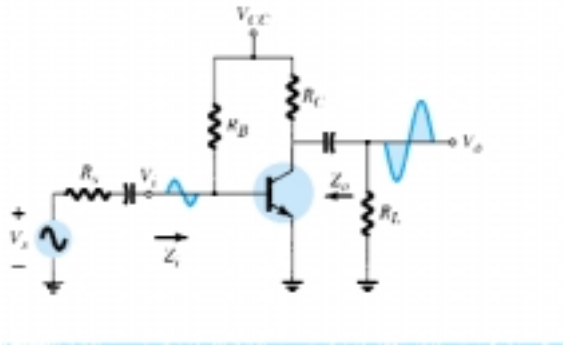
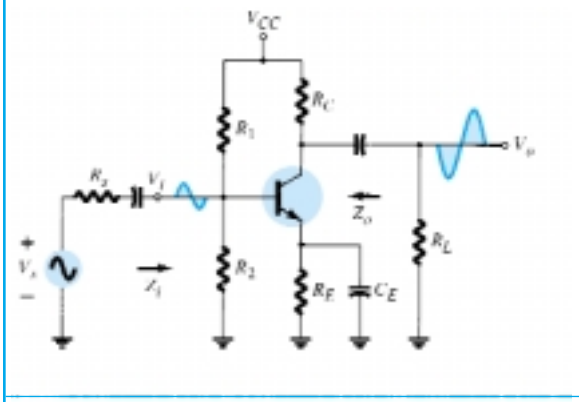
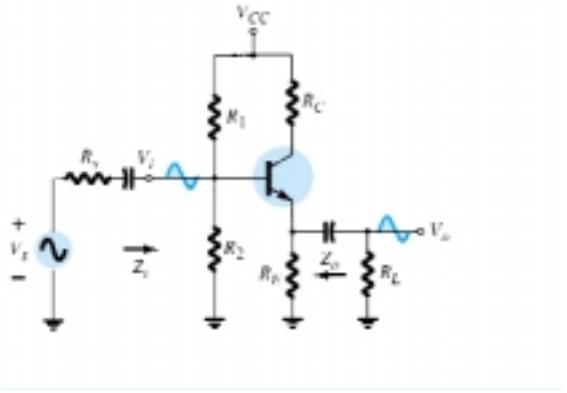
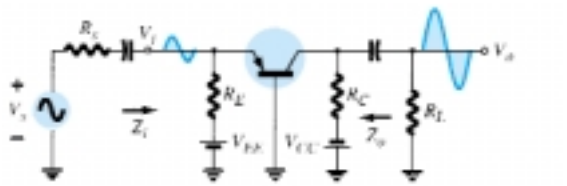
Configuration	$A_v = V_o/V_i$	Z_i	Z_o
	$\dfrac{-R_L\|R_C)}{r_e}$ $\dfrac{-h_{fe}}{h_{ie}}(R_L\|R_C)$ Including r_o: $-\dfrac{(R_L\|R_C\|r_o)}{r_e}$	$R_B\|\beta r_e$ $R_B\|h_{ie}$ $R_B\|\beta r_e$	R_C R_C $R_C\|r_o$
	$\dfrac{-(R_L\|R_C)}{r_e}$ $\dfrac{-h_{fe}}{h_{ie}}(R_L\|R_C)$ Including r_o: $\dfrac{-(R_L\|R_C\|r_o)}{r_e}$	$R_1\|R_2\|\beta r_e$ $R_1\|R_2\|h_{ie}$ $R_1\|R_2\|\beta r_e$	R_C R_C $R_C\|r_o$
	$R'_E = R_L\|R_E$ $\cong 1$ $\cong 1$ Including r_o: $\cong 1$	$R_1\|R_2\|\beta(r_e + R'_E)$ $R_1\|R_2\|(h_{ie} + h_{fe}R'_E)$ $R_1\|R_2\|\beta(r_e + R'_E)$	$R'_s = R_s\|R_1\|R_2$ $R_E\|\left(\dfrac{R'_s}{\beta} + r_e\right)$ $R_E\|\left(\dfrac{R'_s + h_{ie}}{h_{fe}}\right)$ $R_E\|\left(\dfrac{R'_s}{\beta} + r_e\right)$
	$\cong \dfrac{-(R_L\|R_C)}{r_e}$ $\cong \dfrac{-h_{fb}}{h_{ib}}(R_L\|R_C)$ Including r_o: $\cong \dfrac{-(R_L\|R_C\|r_o)}{r_e}$	$R_E\|r_e$ $R_E\|h_{ib}$ $R_E\|r_e$	R_C R_C $R_C\|r_o$

TABLE 10.1 Summary of Transistor Configurations (A_v, Z_i, Z_o) (Continued)

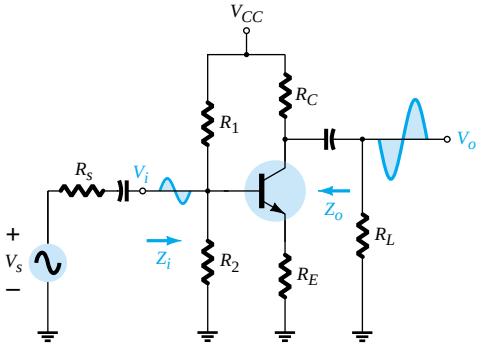
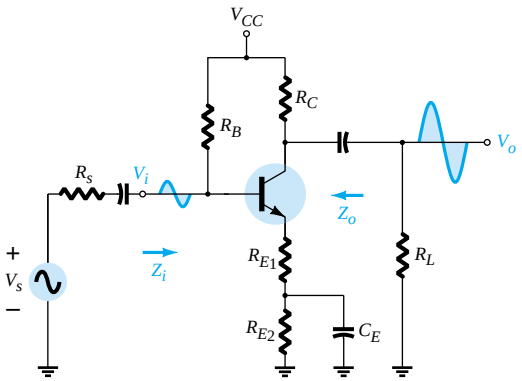
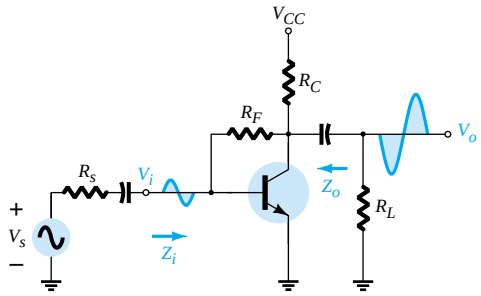
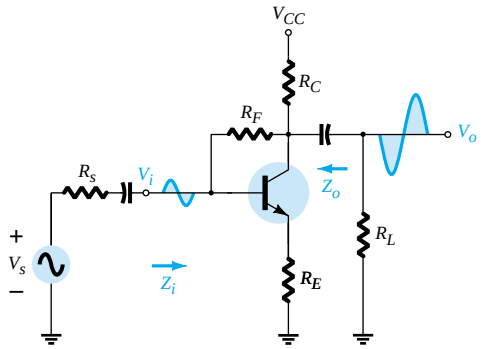
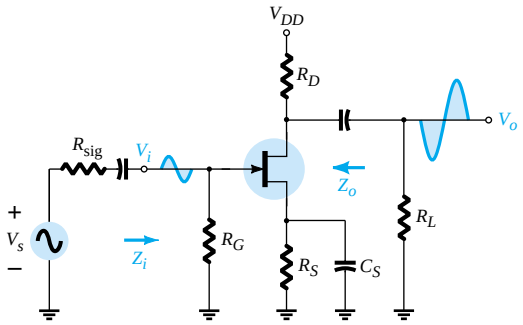
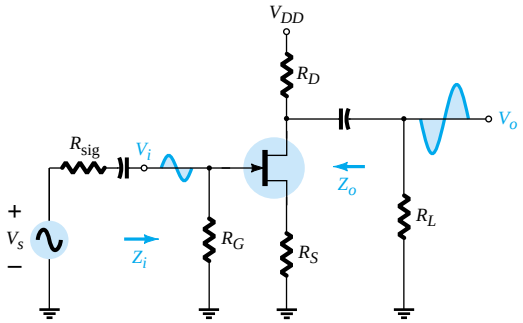
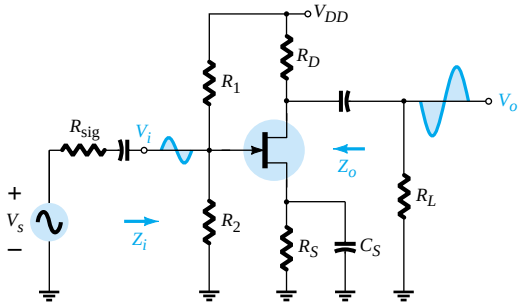
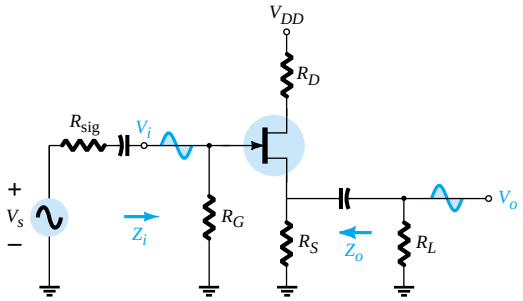
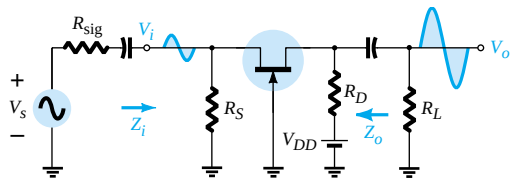
Configuration	$A_v = V_o/V_i$	Z_i	Z_o
	$\frac{-R_L\|R_C)}{R_E}$ $\frac{-(R_L\|R_C)}{R_E}$ Including r_o: $\frac{-(R_L\|R_C)}{R_E}$	$R_1\|R_2\|\beta(r_e + R_E)$ $R_1\|R_2\|(h_{ie} + h_{fe}R_E)$ $R_1\|R_2\|\beta(r_e + R_E)$	R_C R_C $\cong R_C$
	$\frac{-R_L\|R_C}{R_{E_1}}$ $\frac{-(R_L\|R_C)}{R_{E_1}}$ Including r_o: $\frac{-R_L\|R_C}{R_{E_1}}$	$R_B\|\beta(r_e + R_{E_1})$ $R_B\|(h_{ie} + h_{fe}R_{E_1})$ $R_B\|\beta(r_e + R_{E_1})$	R_C R_C $\cong R_C$
	$\frac{-(R_L\|R_C)}{r_e}$ $\frac{-h_{fe}}{h_{ie}}(R_L\|R_C)$ Including r_o: $\frac{-(R_L\|R_C\|r_o)}{r_e}$	$\beta r_e\|\frac{R_F}{\|A_v\|}$ $h_{ie}\|\frac{R_F}{\|A_v\|}$ $\beta r_e\|\frac{R_F}{\|A_v\|}$	R_C R_C $R_C\|R_F\|r_o$
	$\frac{-(R_L\|R_C)}{R_E}$ $\frac{-(R_L\|R_C)}{R_E}$ Including r_o: $\cong \frac{-(R_L\|R_C)}{R_E}$	$\beta R_E\|\frac{R_F}{\|A_v\|}$ $h_{fe}R_E\|\frac{R_F}{\|A_v\|}$ $\cong \beta R_E\|\frac{R_F}{\|A_v\|}$	$\cong R_C\|R_F$ $\cong R_C\|R_F$ $\cong R_C\|R_F$

TABLE 10.1 (Continued)

Configuration	$A_v = V_o/V_i$	Z_i	Z_o
	$-g_m(R_D\|\|R_L)$ Including r_d: $-g_m(R_D\|\|R_L\|\|r_d)$	R_G R_G	R_D $R_D\|\|r_d$
	$\dfrac{-g_m(R_D\|\|R_L)}{1+g_mR_S}$ Including r_d: $\dfrac{-g_m(R_D\|\|R_L)}{1+g_mR_S+\dfrac{R_D+R_S}{r_d}}$	R_G R_G	$\dfrac{R_D}{1+g_mR_S}$ $\cong \dfrac{R_D}{1+g_mR_S}$
	$-g_m(R_D\|\|R_L)$ Including r_d: $-g_m(R_D\|\|R_L\|\|r_d)$	$R_1\|\|R_2$ $R_1\|\|R_2$	R_D $R_D\|\|r_d$
	$\dfrac{g_m(R_S\|\|R_L)}{1+g_m(R_S\|\|R_L)}$ Including r_d: $= \dfrac{g_mr_d(R_S\|\|R_L)}{r_d+R_D+g_mr_d(R_S\|\|R_L)}$	R_G R_G	$R_S\|\|1/g_m$ $\dfrac{R_S}{1+\dfrac{g_mr_dR_S}{r_d+R_D}}$
	$g_m(R_D\|\|R_L)$ Including r_d: $\cong g_m(R_D\|\|R_L)$	$\dfrac{R_S}{1+g_mR_S}$ $Z_i = \dfrac{R_S}{1+\dfrac{g_mr_dR_S}{r_d+R_D\|\|R_L}}$	R_D $R_D\|\|r_d$

10.11 CASCADED SYSTEMS

The two-port systems approach is particularly useful for cascaded systems such as that appearing in Fig. 10.38, where A_{v_1}, A_{v_2}, A_{v_3}, and so on, are the voltage gains of each stage *under loaded conditions.* That is, A_{v1} is determined with the input impedance to A_{v_2} acting as the load on A_{v_1}. For A_{v_2}, A_{v_1} will determine the signal strength and source impedance at the input to A_{v_2}. The total gain of the system is then determined by the product of the individual gains as follows:

$$A_{v_T} = A_{v_1} \cdot A_{v_2} \cdot A_{v_3} \cdots \tag{10.53}$$

and the total current gain by

$$A_{i_T} = -A_{v_T} \frac{Z_{i_1}}{R_L} \tag{10.54}$$

No matter how perfect the system design, the application of a load to a two-port system will affect the voltage gain. Therefore, there is no possibility of a situation where A_{v_1}, A_{v_2}, and so on, of Fig. 10.38 are simply the no-load values. The loading of each succeeding stage must be considered. The no-load parameters can be used to determine the loaded gains of Fig. 10.38, but Eq. (10.53) requires the loaded values.

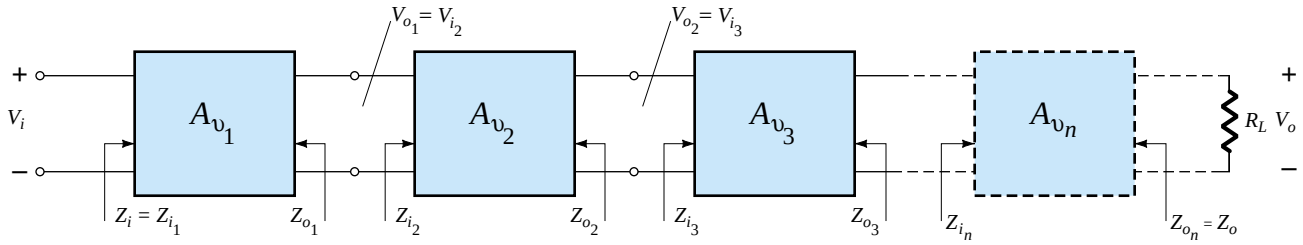


Figure 10.38 Cascaded system.

EXAMPLE 10.9

The two-stage system of Fig. 10.39 employed a transistor emitter-follower configuration prior to a common-base configuration to ensure that the maximum percent of the applied signal appears at the input terminals of the common-base amplifier. In Fig. 10.39, the no-load values are provided for each system, with the exception of Z_i and Z_o for the emitter-follower, which are the loaded values. For the configuration of Fig. 10.39, determine:

(a) The loaded gain for each stage.
(b) The total gain for the system, A_v and A_{v_s}.
(c) The total current gain for the system.
(d) The total gain for the system if the emitter-follower configuration were removed.

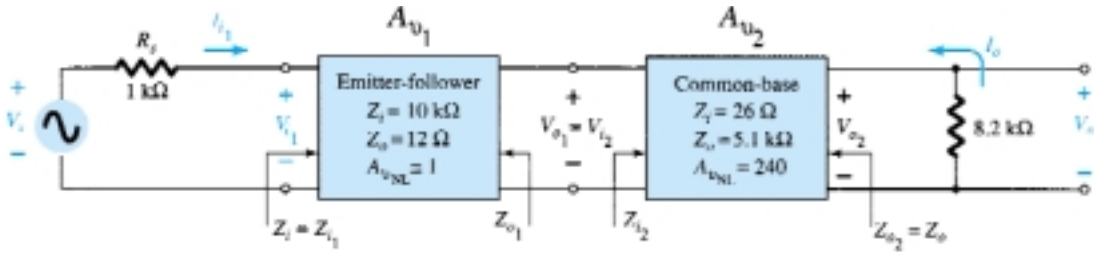


Figure 10.39 Example 10.9.

Solution

(a) For the emitter-follower configuration, the loaded gain is

$$V_{o_1} = \frac{Z_{i_2} A_{v_{NL}} V_{i_1}}{Z_{i_2} + Z_{o_1}} = \frac{(26\ \Omega)(1)V_{i_1}}{26\ \Omega + 12\ \Omega} = 0.684V_{i_1}$$

and $$A_{v_1} = \frac{V_{o_1}}{V_{i_1}} = \mathbf{0.684}$$

For the common-base configuration,

$$V_{o_2} = \frac{R_L A_{v_{NL}} V_{i_2}}{R_L + R_{o_2}} = \frac{(8.2\ \text{k}\Omega)(240)V_{i_2}}{8.2\ \text{k}\Omega + 5.1\ \text{k}\Omega} = 147.97V_{i_2}$$

and $$A_{v_L} = \frac{V_{o_2}}{V_{i_2}} = \mathbf{147.97}$$

(b) $$A_{v_T} = A_{v_1}A_{v_2}$$
$$= (0.684)(147.97)$$
$$= \mathbf{101.20}$$
$$A_{v_s} = \frac{Z_{i_1}}{Z_{i_1} + R_s} A_{v_T} = \frac{(10\ \text{k}\Omega)(101.20)}{10\ \text{k}\Omega + 1\ \text{k}\Omega}$$
$$= \mathbf{92}$$

(c) $$A_{i_T} = -A_{v_T}\frac{Z_{i_1}}{R_L} = -(101.20)\left(\frac{10\ \text{k}\Omega}{8.2\ \text{k}\Omega}\right)$$
$$= \mathbf{-123.41}$$

(d) $$V_{i_{CB}} = \frac{Z_{i_{CB}} V_s}{Z_{i_{CB}} + R_s} = \frac{(26\ \Omega)V_s}{26\ \Omega + 1\ \text{k}\Omega} = 0.025V_s$$

and $\frac{V_i}{V_s} = 0.025$ with $\frac{V_o}{V_i} = 147.97$ from above

and $$A_{v_s} = \frac{V_i}{V_s}\frac{V_o}{V_i} = (0.025)(147.97) = \mathbf{3.7}$$

In total, therefore, the gain is about 25 times greater with the emitter-follower configuration to draw the signal to the amplifier stages. Take note, however, that it was also important that the output impedance of the first stage was relatively close to the input impedance of the second stage or the signal would have been "lost" again by the voltage-divider action.

10.12 PSPICE WINDOWS

Loaded Voltage-Divider BJT Transistor Configuration

The computer analysis of this section includes a PSpice Windows evaluation of the response of a loaded BJT and FET amplifier with a source resistance. The BJT network of Fig. 10.40 employs the same unloaded configuration examined in the PSpice analysis of Chapter 8, where the unloaded gain was 369 (Example 8.2, $r_e = 18.44\ \Omega$). For the transistor, all the parameters listed under **Model Editor** were removed except I_s and beta, which were set to 2E-15A and 90, respectively. In this way, the results

will be as close to the hand-written solutions as possible without going to the controlled source equivalents. Note the placement of the **VPRINT1** option to pick up the voltage lost across the source resistance and to note if there is any drop in gain across the capacitor. The option **Do not auto-run Probe** was chosen, and under **Analysis Setup,** the **AC Sweep** was set at a fixed frequency of 10 kHz. In addition, **Display Results on Schematic** under **Analysis** was chosen and the **Voltage Display** enabled.

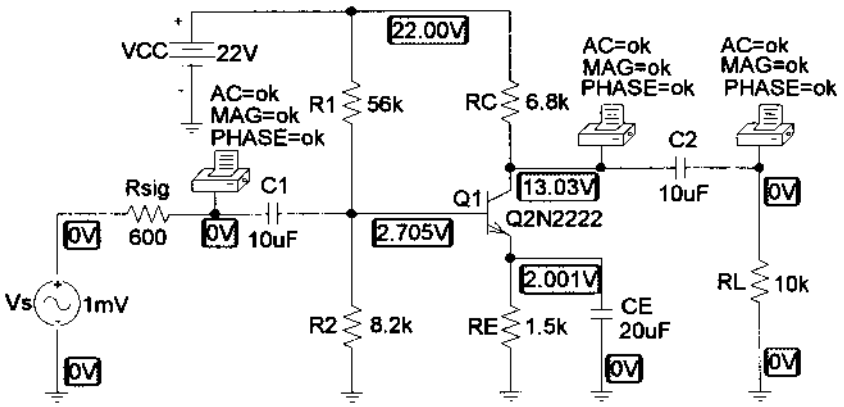


Figure 10.40 Loaded voltage-divider BJT transistor configuration.

An **Analysis** resulted in the dc levels appearing in Fig. 10.40. In particular, note the zero volt levels at the left side of C_1 and the right side of C_2. In addition, note that V_{BE} is essentially 0.7 V and the dc levels of each terminal of the transistor are very close to those calculated in Example 8.2 (using the approximate approach). Reviewing the output file following **Analysis-Examine Output** will result in the data listings of Fig. 10.41. The nodes are defined in the **Schematics Netlist,** and the **BJT MODEL PARAMETERS** reveal our choices for this run—although the last three are default values. The **SMALL-SIGNAL BIAS SOLUTION** simply confirms the levels printed on the schematic, and the **Operating Point Information** reveals that beta (dc and ac) is 90, that V_{BE} is 0.7 V, that I_C is 1.32 mA, and that I_B is 14.7 μA (in addition to a host of other levels). The **AC ANALYSIS** reveals that the voltage on the other side of R_{sig} is about 0.7 mV, resulting in a drop of about 0.3 mV (30% loss in signal voltage) of the applied signal across R_{sig}. The remaining two ac levels are the same, revealing that the capacitor is an effective short circuit for ac. The loaded gain from source to output is 144.9. The gain from the base of the transistor to the output is 144.9 mV/0.7 mV = 207. Both levels are certainly significantly less than the no-load level of 369. If we return to the network and change R_L to 10 MΩ, the output voltage will rise to 243.3 mV, resulting in a gain of 243.3 mV/0.7 mV = 347.57, which is quite close to the hand-calculated, approximate level of 369.

For interest's sake, let us now calculate the loaded voltage gain and compare to the PSpice solution of 144.9.

$$r_e = 18.44\ \Omega$$

and

$$Z_i \cong R_1 \| R_2 \| \beta r_e$$
$$= 56\ \text{k}\Omega \| 8.2\ \text{k}\Omega \| (90)(18.44\ \Omega)$$
$$\cong 1.35\ \text{k}\Omega$$

$$V_i = \frac{Z_i V_s}{Z_i + R_s} = \frac{(1.35\ \text{k}\Omega)V_s}{1.35\ \text{k}\Omega + 0.6\ \text{k}\Omega} = 0.69V_s$$

and

$$\frac{V_i}{V_s} = 0.69$$

```
 ****     CIRCUIT DESCRIPTION
******************************************************************************
* Schematics Netlist *

V_Vs          $N_0001 0  AC 1mV
+SIN 0V 1mV 10kHz 0 0 0
R_Rsig          $N_0001 $N_0002  600

.PRINT          AC
+ VM([$N_0002])
+ VP([$N_0002])
C_C1          $N_0002 $N_0003  10uF
V_VCC          $N_0004 0 22V
R_R1          $N_0004 $N_0003  56k
R_R2          $N_0003 0  8.2k
R_RC          $N_0004 $N_0005  6.8k
R_RE          $N_0006 0  1.5k
C_CE          $N_0006 0  20uF
R_RL          $N_0007 0  10k
C_C2          $N_0005 $N_0007  10uF

.PRINT          AC
+ VM([$N_0005])
+ VP([$N_0005])

.PRINT          AC
+ VM([$N_0007])
+ VP([$N_0007])
Q_Q1          $N_0005 $N_0003 $N_0006 Q2N2222-X

 ****     BJT MODEL PARAMETERS
******************************************************************************

               Q2N2222-X
               NPN
          IS    2.000000E-15
          BF    90
          NF     1
          BR     1
          NR     1

 ****     SMALL SIGNAL BIAS SOLUTION       TEMPERATURE =   27.000 DEG C

******************************************************************************

 NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE

($N_0001)     0.0000                   ($N_0002)     0.0000

($N_0003)     2.7051                   ($N_0004)    22.0000

($N_0005)    13.0280                   ($N_0006)     2.0012

($N_0007)     0.0000

    VOLTAGE SOURCE CURRENTS
    NAME         CURRENT

    V_Vs          0.000E+00
    V_VCC        -1.664E-03

    TOTAL POWER DISSIPATION   3.66E-02  WATTS

 ****     OPERATING POINT INFORMATION      TEMPERATURE =   27.000 DEG C

******************************************************************************

**** BIPOLAR JUNCTION TRANSISTORS

NAME          Q_Q1
MODEL         Q2N2222-X
IB            1.47E-05
IC            1.32E-03
VBE           7.04E-01
VBC          -1.03E+01
VCE           1.10E+01
BETADC        9.00E+01
GM            5.10E-02
RPI           1.76E+03
RX            0.00E+00
RO            1.00E+12
CBE           0.00E+00
CBC           0.00E+00
CJS           0.00E+00
BETAAC        9.00E+01
CBX           0.00E+00
FT            8.12E+17

 ****     AC ANALYSIS                      TEMPERATURE =   27.000 DEG C

******************************************************************************

  FREQ        VM($N_0002) VP($N_0002)

   1.000E+04   7.025E-04  -5.801E-01

 FREQ        VM($N_0005) VP($N_0005)

  1.000E+04   1.449E-01  -1.782E+02

 FREQ        VM($N_0007) VP($N_0007)

  1.000E+04   1.449E-01  -1.782E+02
```

Figure 10.41 Output file for the network of Fig. 10.40.

$$A_v = \frac{V_o}{V_i} = \frac{R_L A_{v_{NL}}}{R_L + R_o} = \frac{(10\ \text{k}\Omega)(-350.4)}{10\ \text{k}\Omega + 6.8\ \text{k}\Omega}$$
$$= -208.57$$

with
$$A_{v_s} = \frac{V_i}{V_s}\frac{V_o}{V_i} = (0.69)(-208.57)$$
$$\cong \mathbf{-144}$$

which is an excellent comparison with the computer solution.

Loaded JFET Self-Bias Transistor Configuration

The network of Fig. 10.42 is a loaded version of the network examined in Chapter 9, which resulted in a no-load gain of 13.3. In the **Model Editor** dialog box, **Beta** was set to 0.625mA/V^2 and **Vto** = −4V. The remaining parameters were left alone to permit a close comparison with the Chapter 9 solution and because they have less effect on the response than for a BJT transistor.

Again, note the effectiveness of the capacitors to block the dc voltages. In addition, note the small voltage at the gate, indicating that the input impedance to the device is in reality not infinite (although for all practical purposes it is an excellent assumption.) Again, the frequency was set to 10 kHz and an Analysis called for without the **Probe** option. The sequence **Analysis-Examine Output** will result in the listing of Fig. 10.43. The **Schematics Netlist** provides a listing of assigned nodes, and the **OPERATING POINT INFORMATION** reveals that the drain current is 3.36 mA, that V_{GS} is −1.71 V, and that g_m is 2.94 mS. The **AC ANALYSIS** reveals that there is negligible drop across either capacitor at this frequency, and the short-circuit equivalency can be assumed. The output voltage is 5.597mV resulting in a loaded gain of 5.597 compared to the unloaded gain of 13.3. Note also that the drop across R_{sig} is negligible due to the high input impedance of the device.

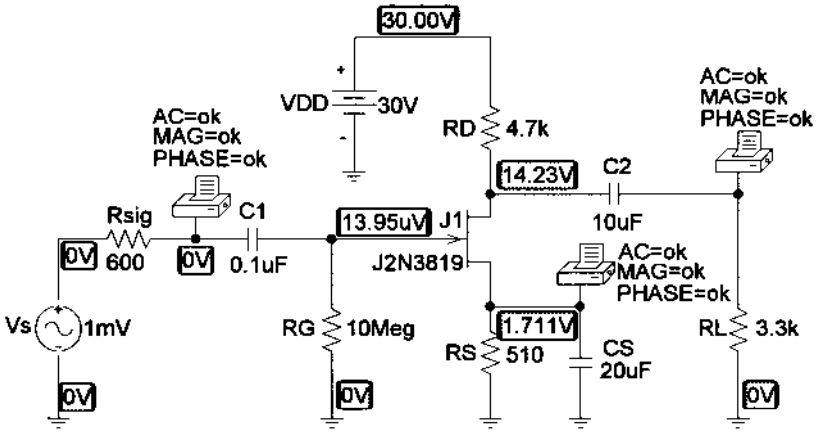


Figure 10.42 Loaded self-bias JFET transistor configuration.

Using the value of g_m hand-calculated earlier, the equation for the loaded gain will result in a gain of 5.62 as shown below—an excellent comparison with the computer solution.

$$A_v = -g_m(R_D \| R_L)$$
$$= -(2.90\ \text{mS})(4.7\ \text{k}\Omega \| 3.3\ \text{k}\Omega)$$
$$= -5.62$$

The results obtained above have clearly substantiated the analysis and equations presented in this chapter for a loaded amplifier.

```
 ****     CIRCUIT DESCRIPTION
******************************************************************************

* Schematics Netlist *

V_Vs          $N_0001 0  AC 1mV
+SIN 0V 1mV 10kHz 0 0 0
R_Rsig         $N_0001 $N_0002  600

.PRINT         AC
+ VM([$N_0002])
+ VP([$N_0002])
C_C1          $N_0002 $N_0003  0.1uF
R_RG          $N_0003 0  10Meg
C_C2          $N_0004 $N_0005  10uF

.PRINT         AC
+ VM([$N_0006])
+ VP([$N_0006])

.PRINT         AC
+ VM([$N_0005])
+ VP([$N_0005])
V_VDD          $N_0007 0 30V
R_RD          $N_0007 $N_0004  4.7k
R_RS          $N_0006 0  510
C_CS          $N_0006 0  20uF
J_J1          $N_0004 $N_0003 $N_0006 J2N3819-X1
R_RL          $N_0005 0  3.3k

 ****     Junction FET MODEL PARAMETERS
******************************************************************************
               J2N3819-X1
               NJF
         VTO   -4
        BETA  625.000000E-06
      LAMBDA    2.250000E-03
          IS   33.570000E-15
         ISR  322.400000E-15
       ALPHA  311.700000E-06
          VK  243.6
          RD    1
          RS    1
         CGD    1.600000E-12
         CGS    2.414000E-12
           M     .3622
       VTOTC   -2.500000E-03
     BETATCE    -.5
          KF    9.882000E-18

 ****     SMALL SIGNAL BIAS SOLUTION       TEMPERATURE =   27.000 DEG C
******************************************************************************
 NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE

($N_0001)    0.0000                   ($N_0002)    0.0000

($N_0003) 13.95E-06                   ($N_0004)   14.2280

($N_0005)    0.0000                   ($N_0006)    1.7114

($N_0007)   30.0000

     VOLTAGE SOURCE CURRENTS
     NAME         CURRENT

     V_Vs          0.000E+00
     V_VDD        -3.356E-03

     TOTAL POWER DISSIPATION   1.01E-01  WATTS
  ****     OPERATING POINT INFORMATION      TEMPERATURE =   27.000 DEG C
******************************************************************************
**** JFETS


NAME          J_J1
MODEL         J2N3819-X1
ID            3.36E-03
VGS          -1.71E+00
VDS           1.25E+01
GM            2.94E-03
GDS           7.34E-06
CGS           1.68E-12
CGD           5.97E-13 *
  ****     AC ANALYSIS                      TEMPERATURE =   27.000 DEG C
******************************************************************************
   FREQ        VM($N_0002) VP($N_0002)

    1.000E+04   9.999E-04  -1.213E-02

   FREQ        VM($N_0006) VP($N_0006)

    1.000E+04   2.297E-06  -8.979E+01

   FREQ        VM($N_0005) VP($N_0005)

    1.000E+04   5.597E-03  -1.799E+02
```

Figure 10.43 Output file for the network of Fig. 10.42.

PROBLEMS

§ 10.3 Effect of a Load Impedance (R_L)

1. For the fixed-bias configuration of Fig. 10.44:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Calculate the gain A_v using the model of part (b) and Eq. (10.3).
(d) Determine the current gain using Eq. (10.6).
(e) Determine A_v, Z_i, and Z_o using the r_e model and compare with the solutions above.

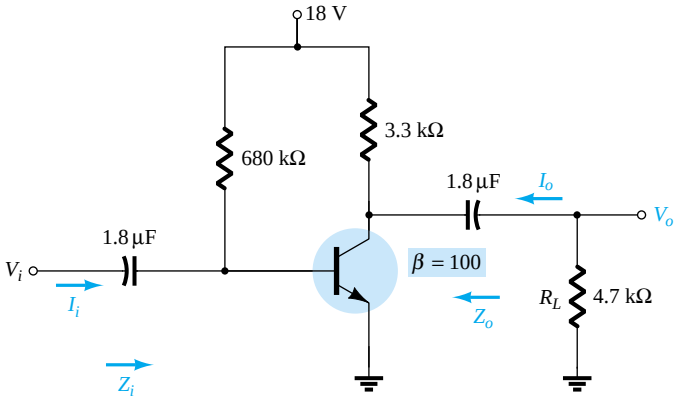


Figure 10.44 Problems 1, 2, and 3

* **2.** (a) Draw the dc and ac load lines for the network of Fig. 10.44 on the characteristics of Fig. 10.45.
(b) Determine the peak-to-peak value of I_c and V_{ce} from the graph if V_i has a peak value of 10 mV. Determine the voltage gain $A_v = V_o/V_i$ and compare with the solution obtained in Problem 1.

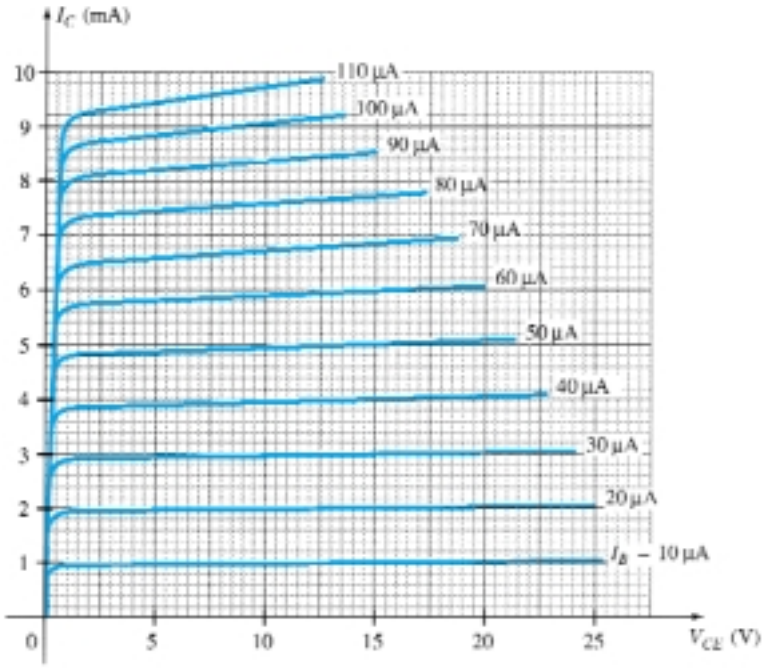


Figure 10.45 Problems 2 and 7

3. (a) Determine the voltage gain A_v for the network of Fig. 10.44 for $R_L = 4.7$, 2.2, and 0.5 kΩ. What is the effect of decreasing levels of R_L on the voltage gain?
(b) How will Z_i, Z_o, and $A_{v_{NL}}$ change with decreasing values of R_L?

§ 10.4 Effect of a Source Impedance (R_s)

* **4.** For the network of Fig. 10.46:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Determine A_v using the results of part (b).
(d) Determine A_{v_s}.
(e) Determine A_{v_s} using the r_e model and compare the results to that obtained in part (d).
(f) Change R_s to 1 kΩ and determine A_v. How does A_v change with the level of R_s?
(g) Change R_s to 1 kΩ and determine A_{v_s}. How does A_{v_s} change with the level of R_s?
(h) Change R_s to 1 kΩ and determine $A_{v_{NL}}$, Z_i, and Z_o. How do they change with change in R_s?

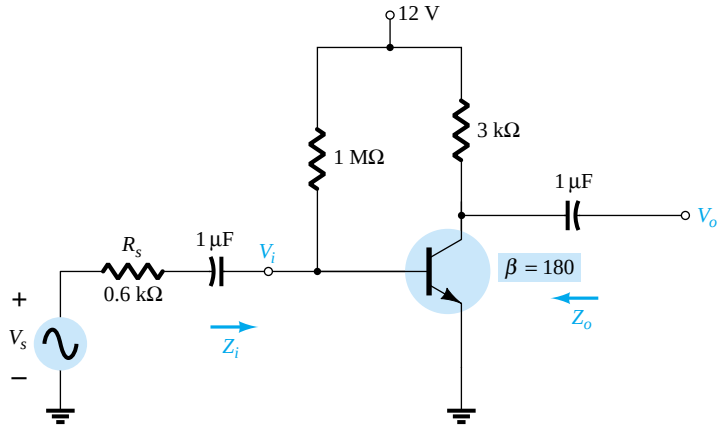


Figure 10.46 Problem 4

§ 10.5 Combined Effect of R_s and R_L

* **5.** For the network of Fig. 10.47:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Determine A_v and A_{v_s}.
(d) Calculate A_i.
(e) Change R_L to 5.6 kΩ and calculate A_{v_s}. What is the effect of increasing levels of R_L on the gain?
(f) Change R_s to 0.5 kΩ (with R_L at 2.7 kΩ) and comment on the effect of reducing R_s on A_{v_s}.
(g) Change R_L to 5.6 kΩ and R_s to 0.5 kΩ and determine the new levels of Z_i and Z_o. How are the impedance parameters affected by changing levels of R_L and R_s?

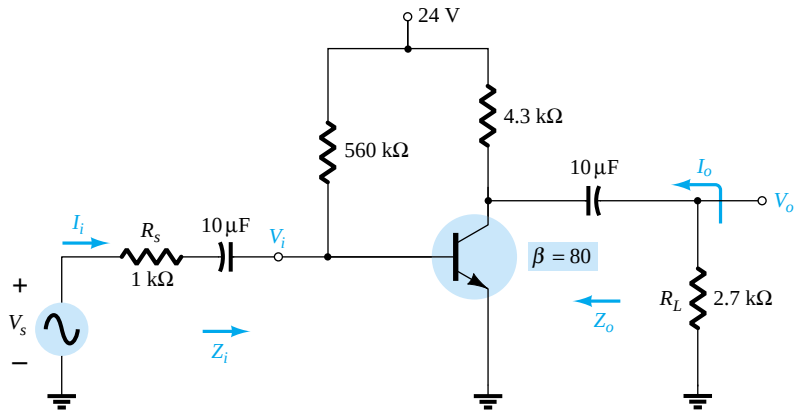


Figure 10.47 Problems 5 and 17

§ 10.6 BJT CE Networks

6. For the voltage-divider configuration of Fig. 10.48:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Calculate the gain A_v using the model of part (b).
(d) Determine the current gain A_i.
(e) Determine A_v, Z_i, and Z_o using the r_e model and compare solutions.

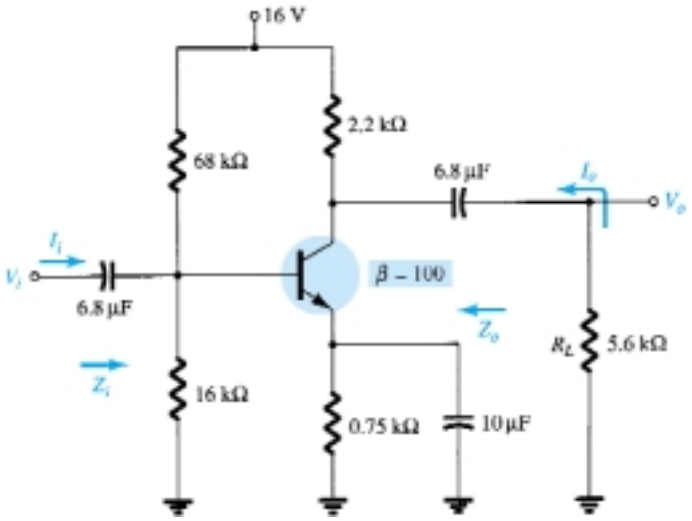


Figure 10.48 Problems 6, 7, and 8

* **7.** (a) Draw the dc and ac load lines for the network of Fig. 10.48 on the characteristics of Fig. 10.45.
(b) Determine the peak-to-peak value of I_c and V_{ce} from the graph if V_i has a peak value of 10 mV. Determine the voltage gain $A_v = V_o/V_i$ and compare the solution with that obtained in Problem 6.

8. (a) Determine the voltage gain A_v for the network of Fig. 10.48 with R_L = 4.7, 2.2, and 0.5 kΩ. What is the effect of decreasing levels of R_L on the voltage gain?
(b) How will Z_i, Z_o, and $A_{v_{NL}}$ change with decreasing levels of R_L?

9. For the emitter-stabilized network of Fig. 10.49:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the values determined in part (a).
(c) Determine A_v and A_{v_s}.
(d) Change R_s to 1 kΩ. What is the effect on $A_{v_{NL}}$, Z_i, and Z_o?
(e) Change R_s to 1 kΩ and determine A_v and A_{v_s}. What is the effect of increasing levels of R_s on A_v and A_{v_s}?

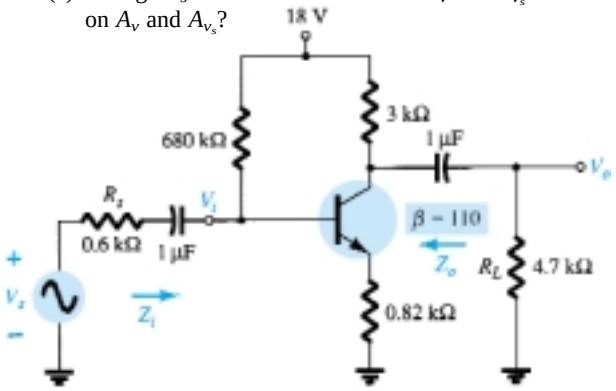


Figure 10.49 Problem 9

§ 10.7 BJT Emitter-Follower Networks

* **10.** For the network of Fig. 10.50:
  (a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
  (b) Sketch the two-port model of Fig. 10.2 with the values determined in part (a).
  (c) Determine A_v and A_{v_s}.
  (d) Change R_s to 1 kΩ and determine A_v and A_{v_s}. What is the effect of increasing levels of R_s on the voltage gains?
  (e) Change R_s to 1 kΩ and determine $A_{v_{NL}}$, Z_i, and Z_o. What is the effect of increasing levels of R_s on the parameters?
  (f) Change R_L to 5.6 kΩ and determine A_v and A_{v_s}. What is the effect of increasing levels of R_L on the voltage gains? Maintain R_s at its original level of 0.6 kΩ.

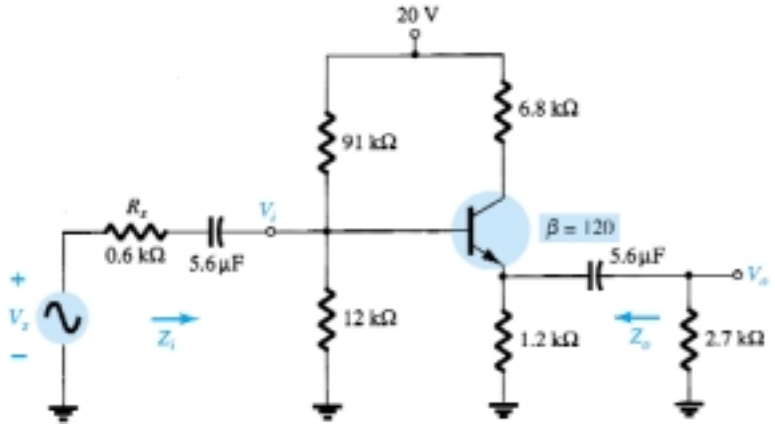


Figure 10.50 Problems 10 and 18

§ 10.8 BJT CB Networks

* **11.** For the common-base network of Fig. 10.51:
  (a) Determine Z_i, Z_o, and $A_{v_{NL}}$.
  (b) Sketch the two-port model of Fig. 10.2 with the parameters of part (a) in place.
  (c) Determine A_v and A_{v_s}.
  (d) Determine A_v and A_{v_s} using the r_e model and compare with the results of part (c).
  (e) Change R_s to 0.5 kΩ and R_L to 2.2 kΩ and calculate A_v and A_{v_s}. What is the effect of changing levels of R_s and R_L on the voltage gains?
  (f) Determine Z_o if R_s changed to 0.5 kΩ with all other parameters as appearing in Fig. 10.51. How is Z_o affected by changing levels of R_s?
  (g) Determine Z_i if R_L is reduced to 2.2 kΩ. What is the effect of changing levels of R_L on the input impedance?

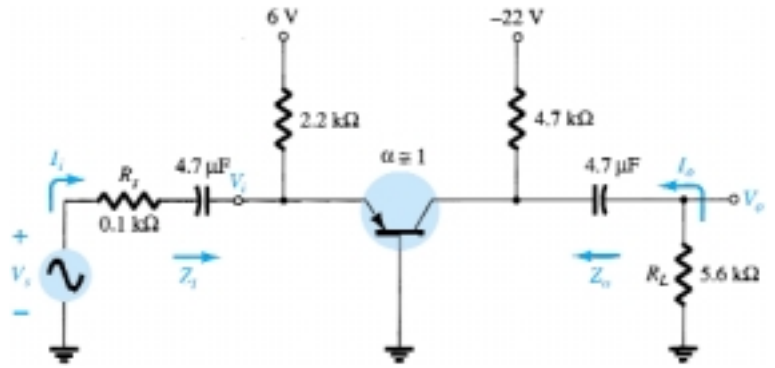


Figure 10.51 Problems 11 and 19

§ 10.9 FET Networks

12. For the self-bias JFET network of Fig. 10.52:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Determine A_v and A_{v_s}.
(d) Change R_L to 6.8 kΩ and R_{sig} to 1 kΩ and calculate the new levels of A_v and A_{v_s}. How are the voltage gains affected by changes in R_{sig} and R_L?
(e) For the same changes as part (d), determine Z_i and Z_o. What was the impact on both impedances?

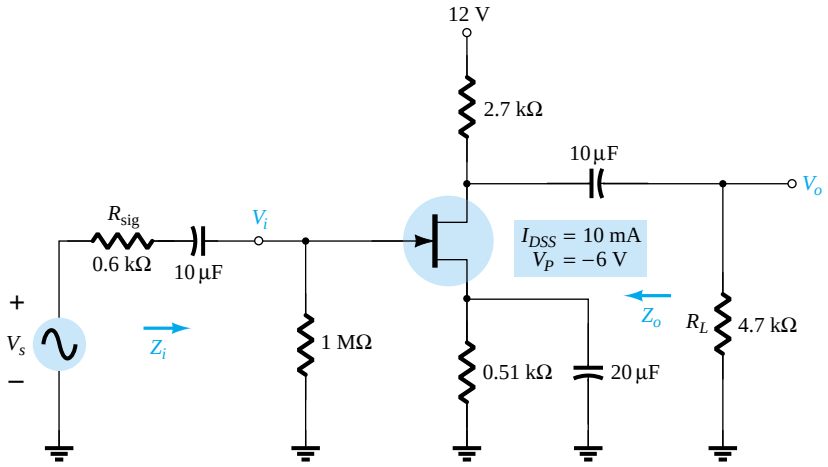


Figure 10.52 Problems 12 and 20

13. For the source-follower network of Fig. 10.53:
(a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
(b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
(c) Determine A_v and A_{v_s}.
(d) Change R_L to 4.7 kΩ and calculate A_v and A_{v_s}. What was the effect of increasing levels of R_L on both voltage gains?
(e) Change R_{sig} to 1 kΩ (with R_L at 2.2 kΩ) and calculate A_v and A_{v_s}. What was the effect of increasing levels of R_{sig} on both voltage gains?
(f) Change R_L to 4.7 kΩ and R_{sig} to 1 kΩ and calculate Z_i and Z_o. What was the effect on both parameters?

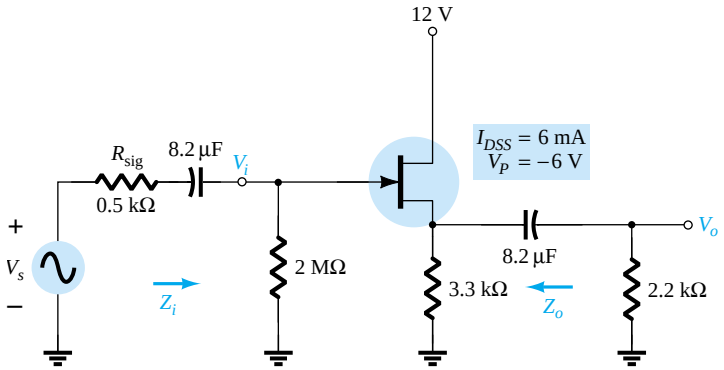


Figure 10.53 Problem 13

* **14.** For the common-gate configuration of Fig. 10.54:
  (a) Determine $A_{v_{NL}}$, Z_i, and Z_o.
  (b) Sketch the two-port model of Fig. 10.2 with the parameters determined in part (a) in place.
  (c) Determine A_v and A_{v_s}.
  (d) Change R_L to 2.2 kΩ and calculate A_v and A_{v_s}. What was the effect of changing R_L on the voltage gains?
  (e) Change R_{sig} to 0.5 kΩ (with R_L at 4.7 kΩ) and calculate A_v and A_{v_s}. What was the effect of changing R_{sig} on the voltage gains?
  (f) Change R_L to 2.2 kΩ and R_{sig} to 0.5 kΩ and calculate Z_i and Z_o. What was the effect on both parameters?

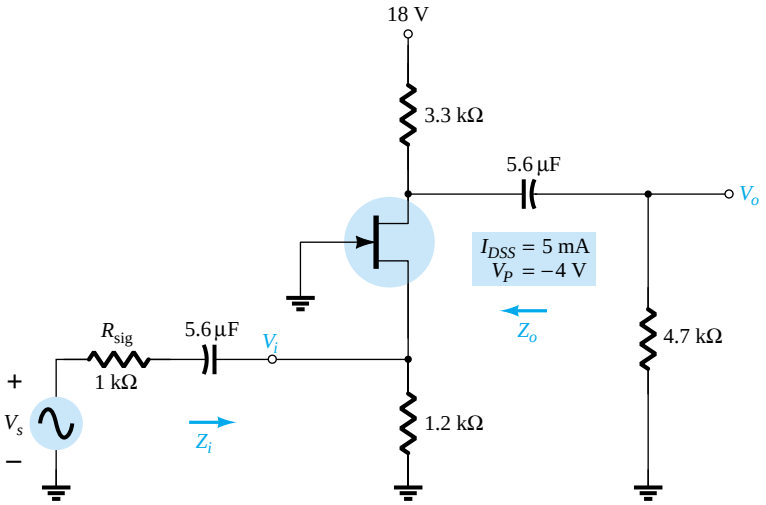


Figure 10.54 Problem 14

§ 10.11 Cascaded Systems

* **15.** For the cascaded system of Fig. 10.55 with two identical stages, determine:
  (a) The loaded voltage gain of each stage.
  (b) The total gain of the system, A_v and A_{v_s}.
  (c) The loaded current gain of each stage.
  (d) The total current gain of the system.
  (e) How Z_i is affected by the second stage and R_L.
  (f) How Z_o is affected by the first stage and R_s.
  (g) The phase relationship between V_o and V_i.

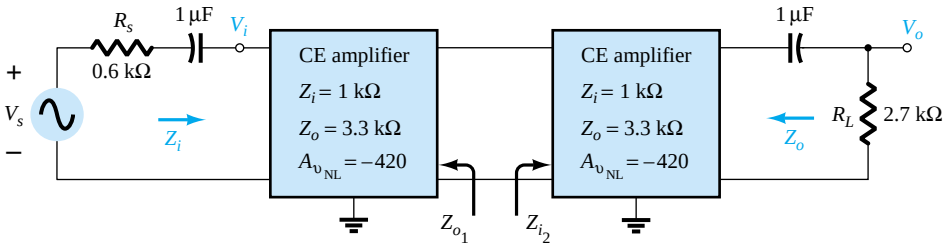


Figure 10.55 Problem 15

* **16.** For the cascaded system of Fig. 10.56, determine:
(a) The loaded voltage gain of each stage.
(b) The total gain of the system, A_v and A_{v_s}.
(c) The loaded current gain of each stage.
(d) The total current gain of the system.
(e) How Z_i is affected by the second stage and R_L.
(f) How Z_o is affected by the first stage and R_s.
(g) The phase relationship between V_o and V_i.

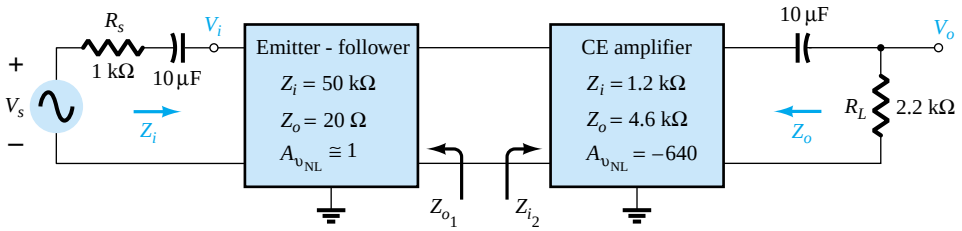


Figure 10.56 Problem 16

§ 10.12 PSpice Windows

17. Using PSpice Windows, determine the level of V_o for $V_s = 1$ mV for the network of Fig. 10.47. For the capacitive elements assume a frequency of 1 kHz.

18. Repeat Problem 17 for the network of Fig. 10.50 and compare the results with those of Problem 10.

19. Repeat Problem 17 for the network of Fig. 10.51 and compare with the results of Problem 11.

20. Repeat Problem 17 for the network of Fig. 10.52 and compare with the results of Problem 12.

*Please Note: Asterisks indicate more difficult problems.